



AKTU

Soft Skills

B.Tech 1st Year

Environment & Ecology

ONE SHOT Revision

Unit-1

Live @ 3:00 PM

Environment & Ecosystem

Soft Skills Playlist in Description/Pin Comment



by M S Tomar Sir

Environment and Ecology

AKTU Syllabus : Unit-1

Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness.

Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem. Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on the Environment, Environmental Impact Assessment, Sustainable Development.

Evs Model Papers attached in App/Telegram

Environment and Ecology

AKTU Syllabus

Unit-2 : Natural Resources

Natural Resources: Introduction, Classification.

Water Resources; Availability, sources and Quality Aspects, Water Borne and Water Induced Diseases, Fluoride and Arsenic Problems in Drinking Water.

Mineral Resources; Material Cycles; Carbon, Nitrogen and Sulfur cycles.

Energy Resources; Conventional and Non conventional Sources of Energy.

Forest Resources; Availability, Depletion of Forests, Environment impact of forest depletion on society.

Environment and Ecology

AKTU Syllabus : Unit-3

Environment: Pollution and their Effects; Public Health Aspects of Environmental

- Water Pollution
- Air Pollution
- Soil Pollution
- Noise Pollution
- Solid waste management

Environment and Ecology

AKTU Syllabus : Unit-4

Current Environmental Issues;

- Global Warming
- Green House Effects
- Climate Change
- Acid Rain
- Ozone Layer Formation and Depletion
- Population Growth and Automobile pollution
- Burning of paddy straw

Environment and Ecology

AKTU Syllabus : Unit-5

Environmental Protection:-

- Environmental Protection Act 1986
- Initiatives by Non Governmental Organizations (NGO's)

Human Population and the Environment:-

- Population growth
- Environmental Education
- Women Education

Environment and Ecology

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Environment and Ecology

UNIT-1 AKTU PYQs

AKTU 2023 – 24 Odd Sem(1st Sem)

Q.1	<i>Who is the father of ecology in India?</i>	2 Marks
Q.2	<i>Differentiate between producers and decomposer with examples.</i>	2 Marks
Q.3	<i>Describe environmental impact assessment (EIA) process.</i>	7 Marks
Q.4	<i>Why are there less number of organisms at the top of the ecological pyramid? Also Write the limitations of ecological pyramid.</i>	7 Marks
Q.5	<i>How you can achieve sustainable development? Also write the reasons of environmental crisis.</i>	7 Marks

UNIT-1 AKTU PYQs

AKTU 2022 – 23 Even Sem(2nd sem)

Q.1	<i>What do you mean by the term Environment? Name the component and segments of environment.</i>	2 Marks
Q.2	<i>What do you mean by the term Ecology? Define the term Producer, Consumers and Decomposers.</i>	2 Marks
Q.3	<i>What is Sustainable development?</i>	2 Marks
Q.4	<i>Define the term Ecological Pyramids. Giving suitable examples, explain the term upright and inverted ecological pyramids.</i>	7 Marks
Q.5	<i>What is the concept of Food Chain and Food Web? Explain different types of food chain and food web stating suitable examples and illustrations.</i>	7 Marks
Q.6	<i>Discuss in details about the term Environmental Impact Assessment (EIA). Enumerate and describe the various steps involved in the EIA process.</i>	7 Marks

Environment and Ecology

UNIT-5 AKTU PYQs

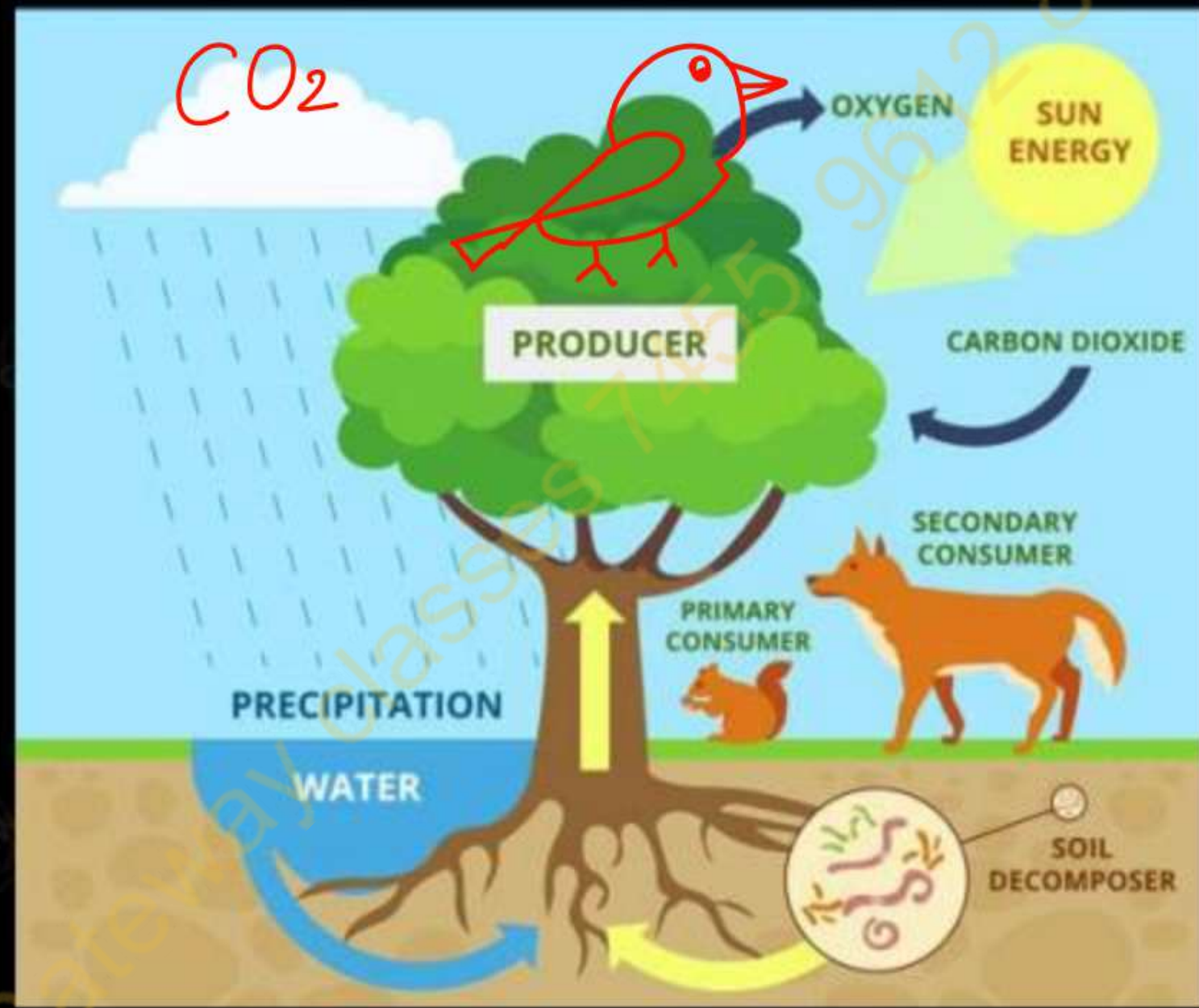
AKTU 2022 – 23 Odd Sem(1st Sem)

Q.1	<i>With the help of flow chart, describe the EIA process.</i>	2 Marks
Q.2	<i>Briefly discuss the effects of industry on environment.</i>	2 Marks
Q.3	<i>Define ecology and ecosystems. What are biotic and abiotic components of an ecosystem?</i>	7 Marks
Q.4	<i>What is an environment? How would you broadly divide the major layers of the atmosphere? State their respective altitude and temperature ranges.</i>	7 Marks
Q.5	<i>Critically examine the role of 'chemical fertilizers' and 'pesticides' use in agriculture.</i>	7 Marks

Environment and Ecology

Non-Living (Abiotic)

- Sunlight
- CO_2
- Water
- Soil



Living (Biotic)

- Zebra
- Bird
- Squirrel
- Plants



"Whatever we see in our surrounding is called environment"

- **Environment** is the surrounding conditions that affect the growth, development, and survival of living organisms.
- It includes both the natural and man-made surroundings, including air, water, food, sunlight, land, plants, animals, and human beings, road buildings, bridges, etc.
- The environment is our basic life support system.

Ecology ("पारिस्थितिकी")

Q. Who is the father of ecology in India? **AKTU-2034-24 Odd Sem**
Ans. Ramdeo Misra

- Ecology is the study of how living things interact with each other and with their environment.
- It is the relationships between plants, animals, and their surroundings.

Types of Environment

1. Natural Environment ("प्राकृतिक पर्यावरण")

"The natural environment is the surroundings or conditions created by nature without significant human intervention".

- Air
- Water
- Soil
- Land
- Forest
- Wild life

2. Man-made Environment (or Built Environment) ("कृत्रिम पर्यावरण")

"The man-made environment is the surroundings or conditions that have been created by human activity".

- Housing
- Energy Sources: Hydro, Thermal, Nuclear, Atomic
- Transportation
- Agriculture
- Technology

Components of Environment ("पर्यावरण के घटक")

Natural Environment

Human Environment

Biotic(Living)
(जीवित)

Abiotic(Non-living)
(निर्जीव)

Activities, Creations
and interaction among
human beings.

The world of living
organisms

e.g.- Plants and Animals

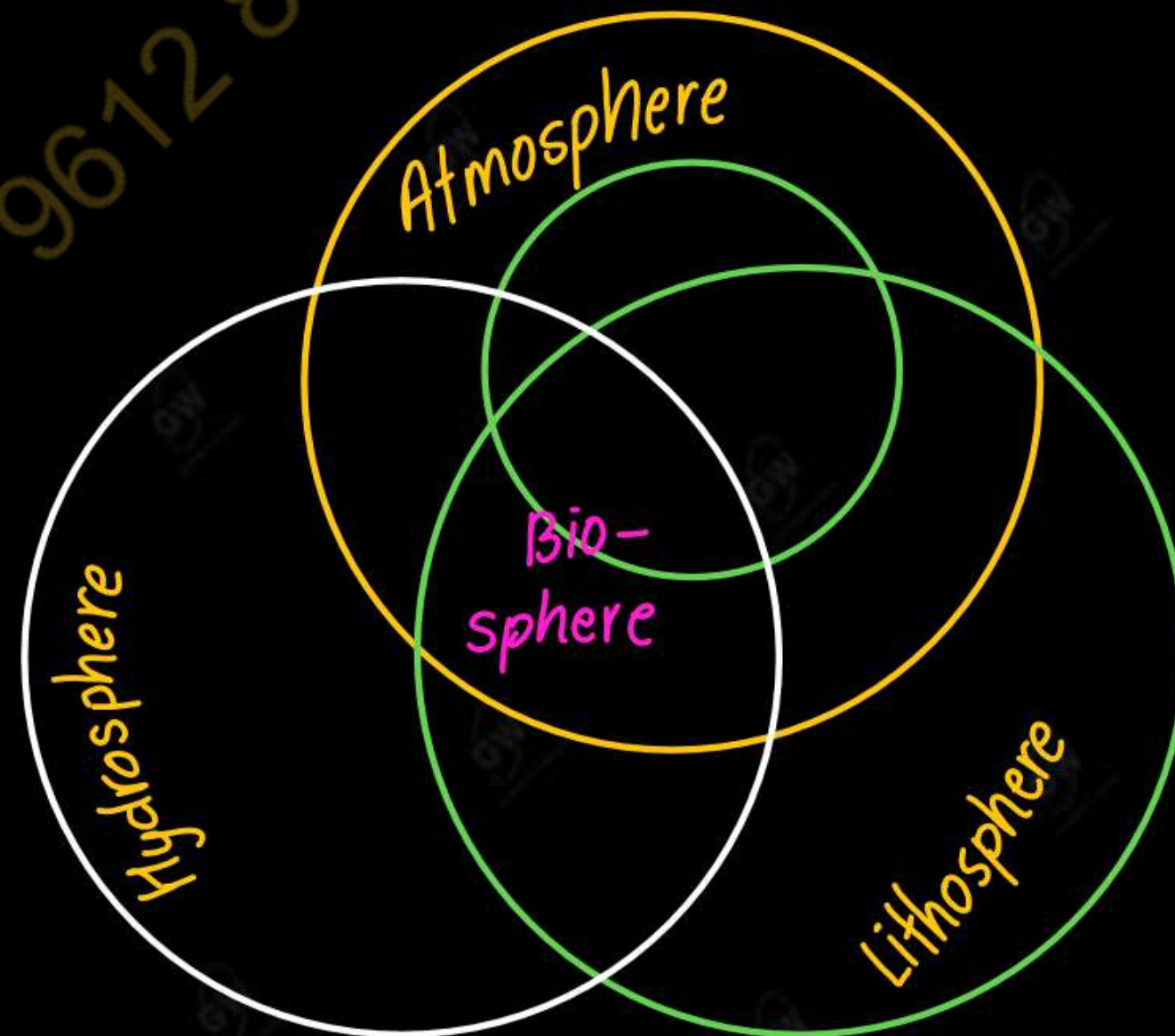
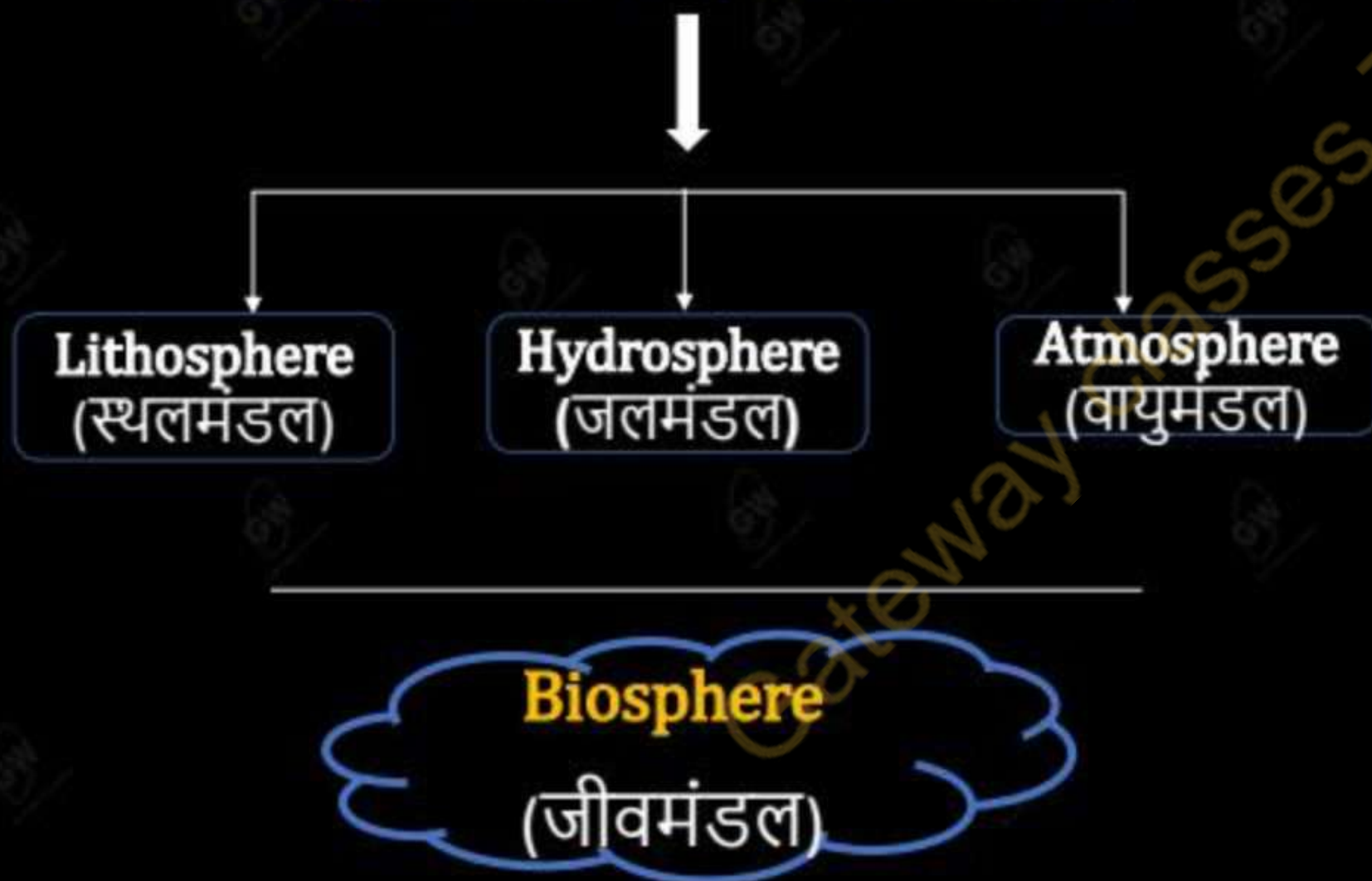
The world of non-living
elements.

e.g. — air, water, Land

The Environment consists of various segments such as

→ Lithosphere, Hydrosphere, Atmosphere and Biosphere.

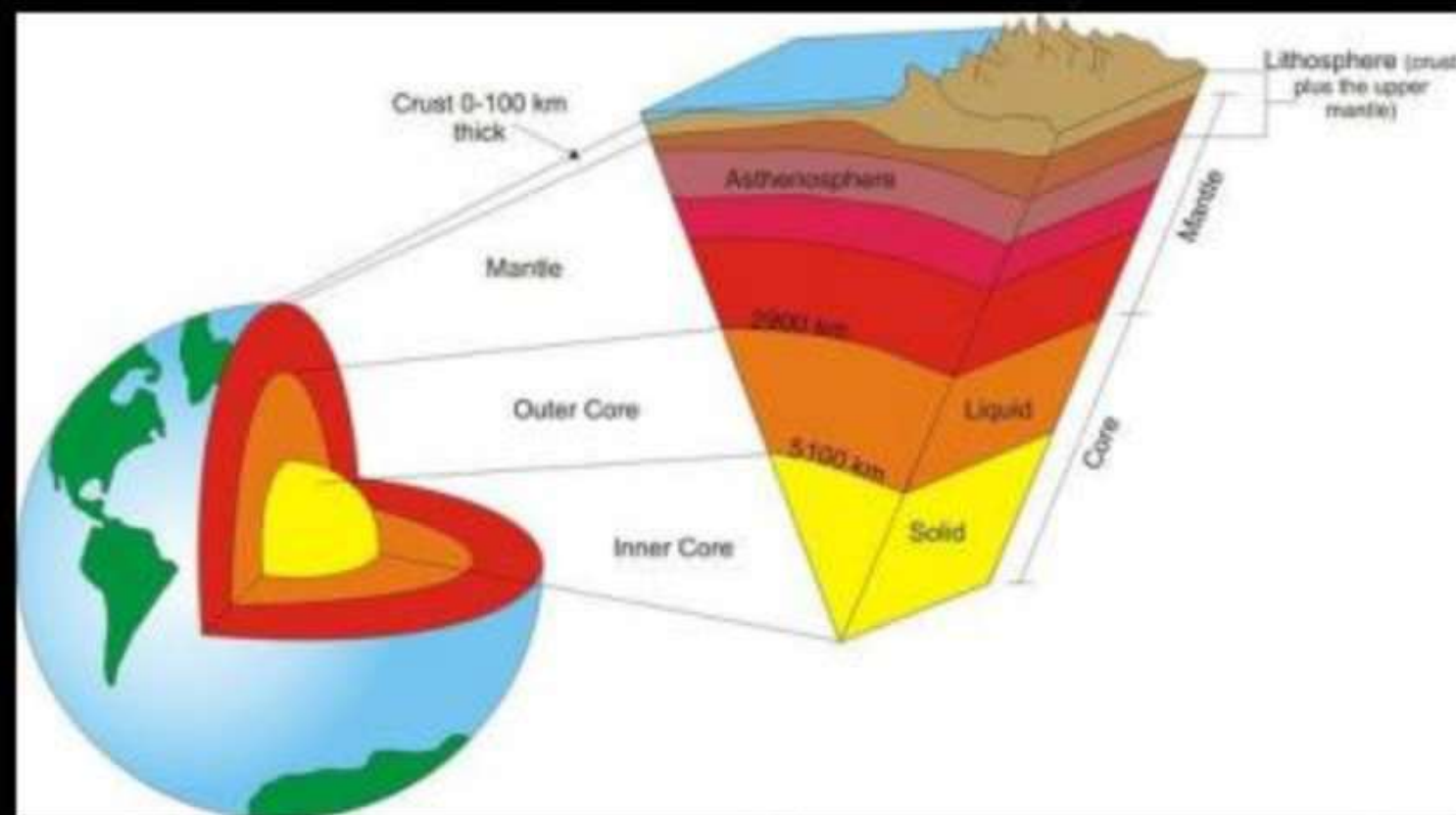
Segments of environment



Domains of the Environment

Lithosphere (स्थलमंडल)

- Solid crust on the hard top layer of the earth.
- Made up of rocks and minerals and covered by thin layer of soil.
- Lithosphere consist of landforms such as mountains, plateaus, plains, valleys, etc.
- Lithosphere is a domain that provides us forests, grasslands of grazing, land for agricultures and human settlements. we also get minerals from lithosphere.



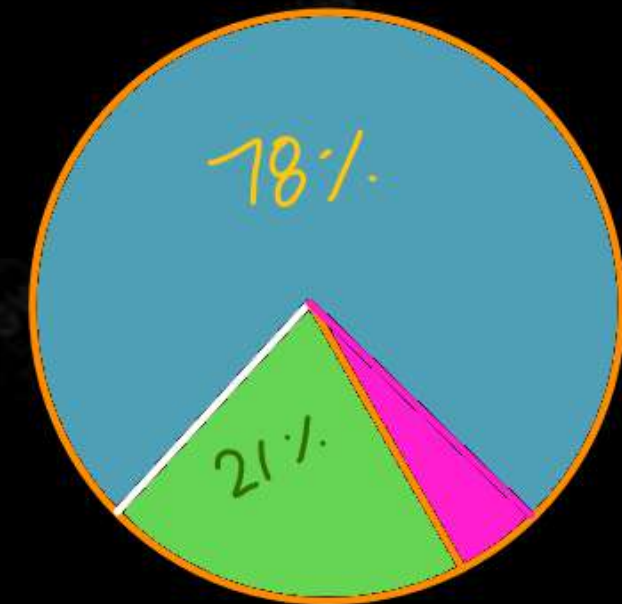
Hydrosphere (जलमंडल)

Segments of environment

- The domain of water = Hydrosphere.
- It comprises various sources of water and different types of water bodies. e.g. – Rivers, lakes, seas, oceans etc.
- It is essential for all living organisms.

Atmosphere (वायुमंडल)

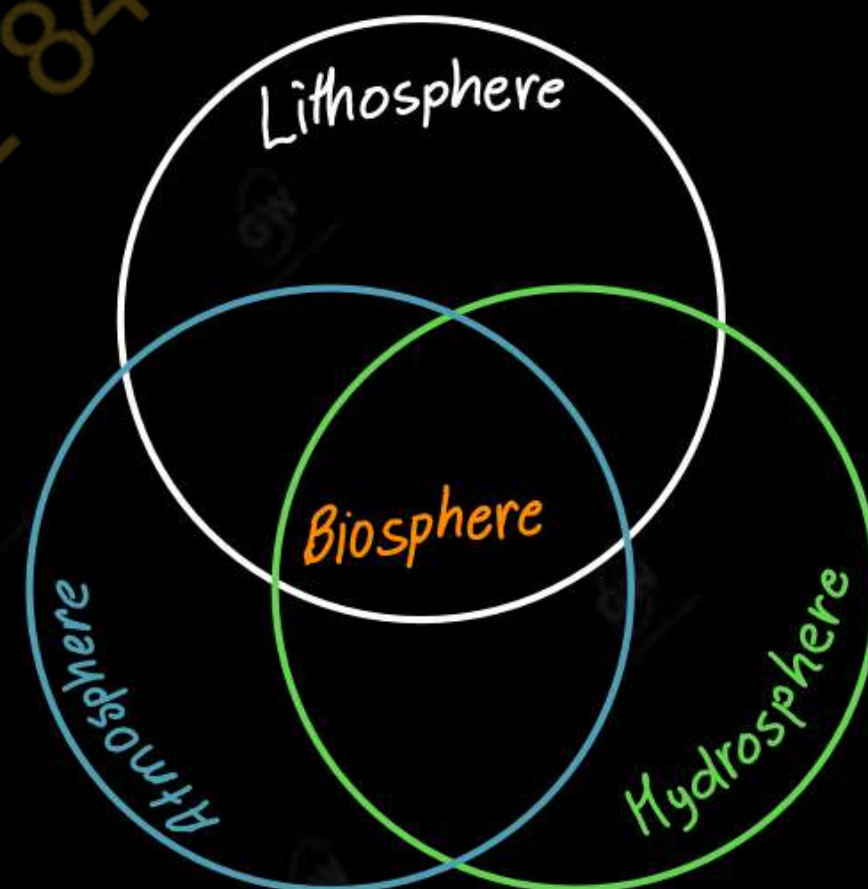
- The atmosphere is the layer of gases surrounding earth that supports life.
- The gravitational force of the earth holds the atmosphere around it.
- It protect us from the harmful rays and heat of the sun.
- It consists of number of gases, dust and water vapour.
- Change in atmosphere = change in weather and climate.



- Nitrogen-78%
- Oxygen-21%
- other gases

Biosphere (जीवमंडल)

- The biosphere is the part of the earth in which life exists.
- It is dependent on the atmosphere(air), hydrosphere(water) and lithosphere(land).
- This denotes the relating of living organism and their interactions with the environment.



Gateway classes 7455 961284

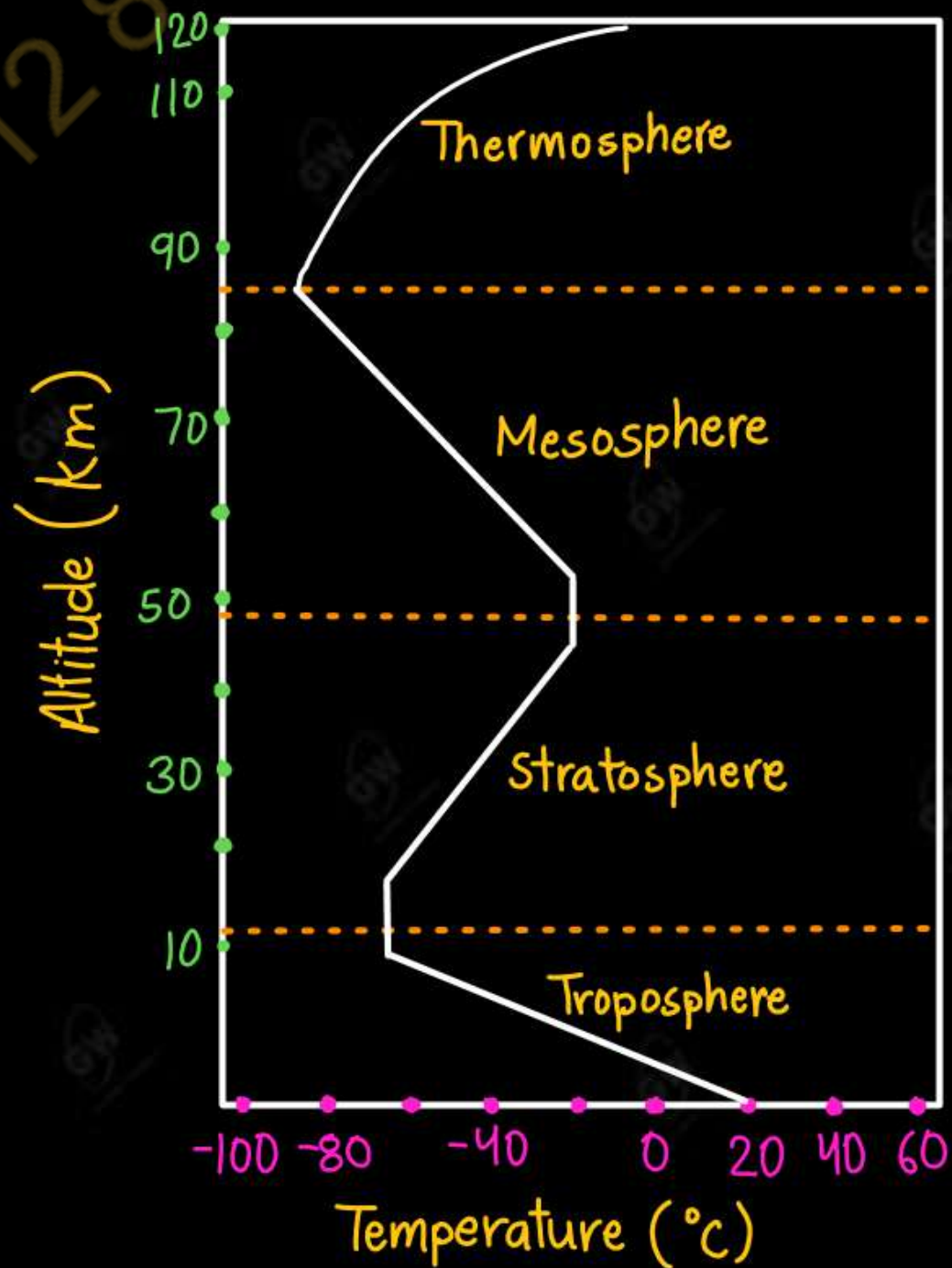
Classification of Atmosphere (वायुमंडल का वर्गीकरण)

AKTU 2014-15, 2022-23

→ On the basis of temperature variation, atmosphere is divided into four major layers – Troposphere, Stratosphere, Mesosphere and Thermosphere.

1. Troposphere (क्षोभमंडल)

- It is the lower portion of the atmosphere, extending upto about 8 km at the poles and 16 km at the equator.
- This is the layer in which most living things exist.
- There is a uniform decrease in temperature with increase in altitude (about $6^{\circ}\text{C}/\text{Km}$) to a minimum of -50°C or -60°C .



2. Stratosphere (समतापमंडल)

- It is above the troposphere; where the temperature is nearly constant upward to about 20 km and then increases upto a maximum of 0°C near its outer limit.
- It contains Ozone which acts as protective layer for the living organisms because it absorbs harmful Ultra Violet (UV) radiations coming from the sun.
- Therefore it is also known as **Ozonosphere**.

3. Mesosphere (मध्यमंडल)

- The Mesosphere lies between the thermosphere and the stratosphere.
- "Meso" means middle, and this is the highest layer of the atmosphere in which the gases are all mixed up rather than being layered by their mass.
- In this layer, the temperature decreases slowly with the altitude but then sharply to a minimum of about -75°C at 80 km.
- They experience increasing friction.

4. Thermosphere (तापमंडल)

- The thermosphere lies between the exosphere and the mesosphere.
- "Thermo" means heat, and the temperature in this layer again rises to a very high value and approaches 2000°C and even more at about 500 km depending upon solar activity.

5. Exosphere

- "Exo" means outside.
- The exosphere is the outermost layer of the Earth's atmosphere.
- It starts at an altitude of about 500 km and goes out to about 10,000 km.
- The exosphere is largely composed of light gases such as hydrogen, carbon dioxide, atomic oxygen, and helium.

Environmental science is a multidisciplinary field that integrates various scientific disciplines to understand and address environmental problems.

Biology: Studies the **living organisms** and their interactions with the environment. This includes understanding ecosystems, biodiversity, and conservation biology.

Chemistry: Focuses on the **chemical processes and substances that impact the environment**. Environmental chemistry involves studying pollutants, their effects, and how to manage them.

Physics: Explores the **physical processes** that occur in the environment, such as **energy transfer, climate change**, and pollution dispersion.

Geology: Examines the **Earth's physical structure and processes**, including soil formation, natural resources, and the impact of geological activities on the environment.

Geography: Analyzes the spatial aspects of the environment, including land use, resource distribution, and the impact of human activities on different regions.

Ecology: Focuses on the relationships between organisms and their environments, studying ecosystems, food chains, and the balance of natural habitats.

Social Sciences: Includes sociology, economics, and political science to understand the human dimensions of environmental issues. This involves studying human behavior, policy-making, and economic impacts on the environment.

Environmental Engineering: Applies engineering principles to develop solutions for environmental problems, such as pollution control, waste management, and sustainable design.

Meteorology: Studies weather and climate patterns, helping to understand and predict climate change and its impacts.

Toxicology: Examines the effects of toxic substances on living organisms and the environment, contributing to risk assessments and the development of safety regulations.

Scope and importance of environment (पर्यावरण का क्षेत्र और महत्व)

1. Ecological Balance

- Maintenance of the equilibrium between living organisms and their environment.
- Supports biodiversity, ensures the survival of species, and maintains natural processes.

2. Human Health

- The impact of the environment on physical and mental well-being.
- Clean air, water, and soil are essential for good health. Pollution can lead to diseases.

3. Economic Importance

- The role of natural resources and ecosystem services in the economy.
- Provides raw materials, supports industries like agriculture, tourism, and fisheries, and offers ecosystem services like pollination and climate regulation.

4. Cultural and Aesthetic Value

- The significance of the environment in cultural heritage, recreation, and aesthetics.
- Natural landscapes inspire art, provide recreational opportunities, and hold cultural significance.

5. Sustainability

- The capacity to maintain ecological and human systems over the long term.
- Ensures resources are available for future generations, promotes conservation, and supports sustainable development.

1. Environmental Education

- Integrating environmental topics in school curriculums.
- Community education programs.

2. Policy and Legislation

- Understanding and supporting environmental laws.
- Active participation in policy-making processes.

3. Sustainable Practices

- Encouraging recycling, waste reduction, and sustainable consumption.
- Promoting renewable energy sources.

4. Global Cooperation

- Importance of international agreements (e.g., Paris Agreement).
- Role of global organizations (e.g., UNEP).

5. Individual Actions

- Personal responsibility in reducing carbon footprints.
- Conservation efforts at the household level.

MODES OF AWARENESS

There are the following ways for awareness.

- Television, Radio, Newspapers, Films, posters, Youtube, etc.
- By organizing seminars/webinars and workshops.
- By including Environmental Science as a subject in the syllabus of School, Colleges and Universities.
- By educating people in their local language.
- Government can introduce Environmental education in rural development programs, agriculture
- Ultimately, we should not forget that preservation of environment is most necessary for survival of living beings on the earth.

Ecosystem

(पर्यावरण तंत्र)

"The system resulting from the integration of all living and non-living factors of the environment".

"A system formed by the interaction of all living organisms with each other and with physical and chemical factors of the environment in which they live, all linked by the transfer of energy and material".

→ All plants, animals, and human beings → Depends on their immediate surroundings.

↓
Relation between the living organisms and their surroundings forms an ecosystem.

Ecosystem

→ Example

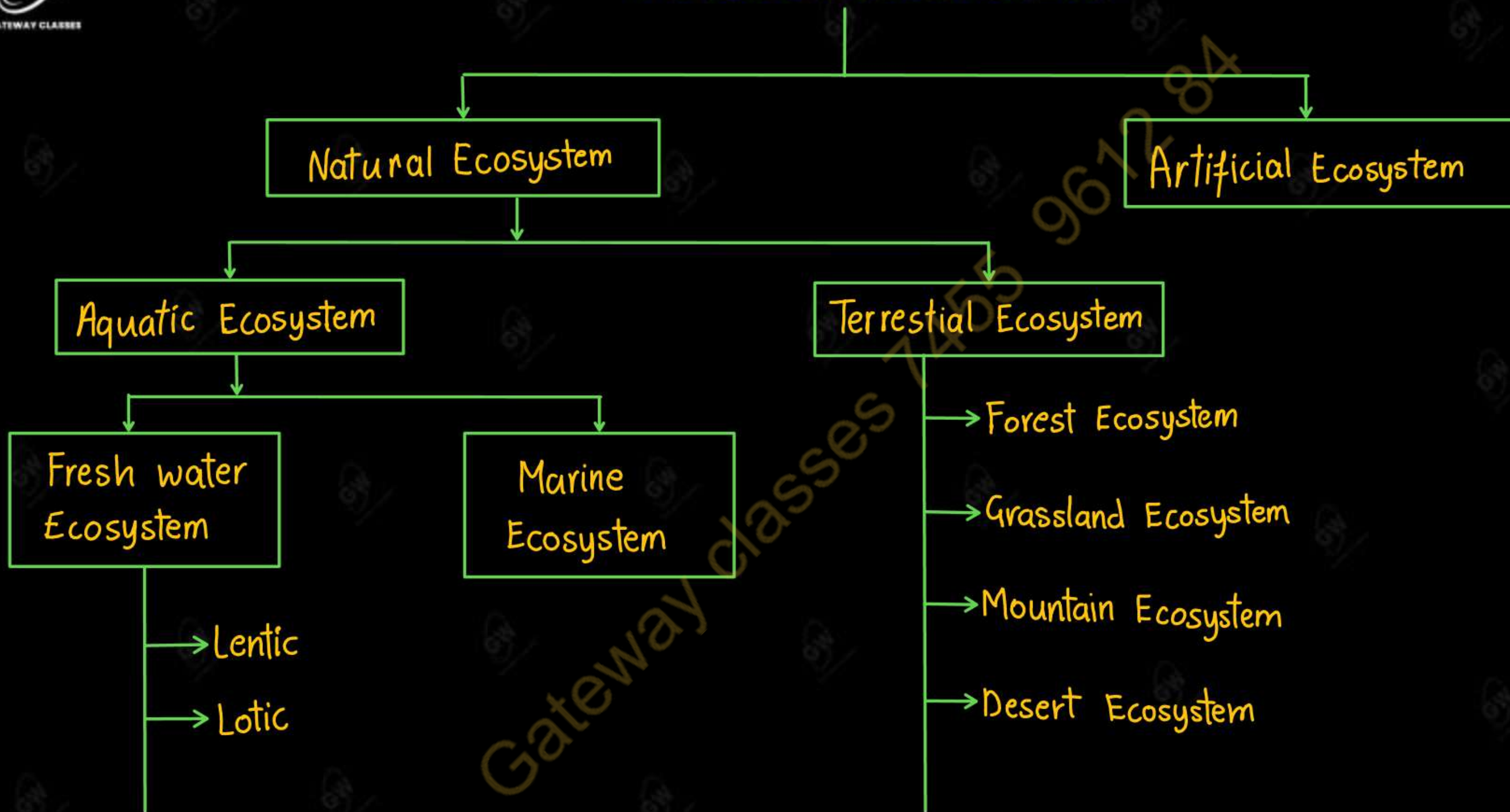
→ Rain, forest, Grassland, Dessert, Lake, Pond etc.

Ecology

→ Study of ecosystem

→ A branch of science which concerned with the inter-relationship of organism and their environment

TYPES OF ECOSYSTEM



TYPES OF ECOSYSTEM

Ecosystem are of two types -

(A) Natural Ecosystems - These operate by themselves under natural conditions without any major interference by man. These are further divided as-

a) Terrestrial Ecosystem - Found on Land : Forest, Grassland, Desert ecosystem etc.

b) Aquatic Ecosystem - It is divided into –

(i) Fresh Water - It may be **Lotic** (Running water as spring, stream or rivers) or **Lentic** (Standing water as lake, ponds, pools etc).

(ii) Marine Water - Like Ocean and seas.

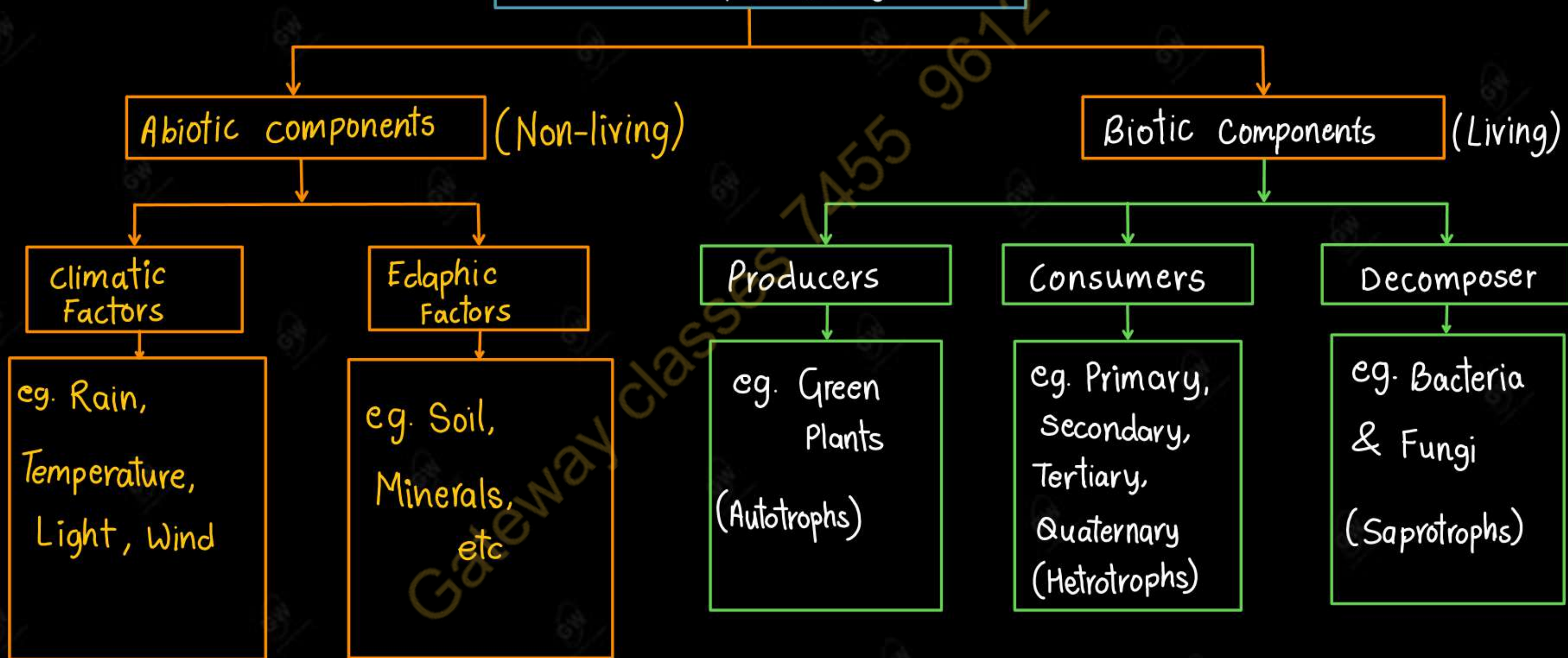
(B) Artificial Ecosystem - It is the ecosystem in which conditions are modified by man.

Examples- Aquarium, Zoo, Cultivated land, Fish farms, Canals, Gardens and Industrial & Urban area.

STRUCTURE OF ECOSYSTEM

(पारिस्थितिकी तंत्र की संरचना)

Component of an Ecosystem



STRUCTURE OF ECOSYSTEM

→ An ecosystem may be divided into two components.

1. Abiotic Components (Non Living)

→ These are the non living components of an ecosystem includes both physical and chemical features.

→ Physical features includes wind, soil, moisture, temperature, light etc. **while** chemical features includes water, gases as oxygen, minerals as iron, sulfur, carbon, nitrogen etc.

→ The kind of chemicals which are present in any ecosystem may regulate the activities of autotrophs and heterotrophs.

2. Biotic Components (Living) - The biotic components of an ecosystem may be divided as

(A) Autotrophs or Producers (स्वपोषी या उत्पादक)

→ These are the living components of an ecosystem which can prepare their own food.

→ These are mainly green plants which prepare their food in presence of sunlight by making use of carbon dioxide, water and chlorophyll through the process of photosynthesis.

STRUCTURE OF ECOSYSTEM

(B) Heterotrophs Consumers (उपभोक्त)

→ These are organisms which feed upon green plants (autotrophs). They are divided as herbivores, carnivores and omnivores.

(i) **Herbivores** (शाकाहारी) – Those organisms which feed on green plants only are called herbivores.

→ They are also called as **primary consumers**.

→ **Examples** are Cow, Buffalo, Goat, Deer, Camel etc.

(ii) **Carnivores** (मांसाहारी) - Those organisms which feed on meat are called carnivores.

→ They are also called as **secondary consumers**.

→ **Examples** are Lion, Tiger, Wolf etc.

(iii) **Omnivores** (सर्वाहारी) - Those organisms which feed on green plants as well as meat are called omnivores.

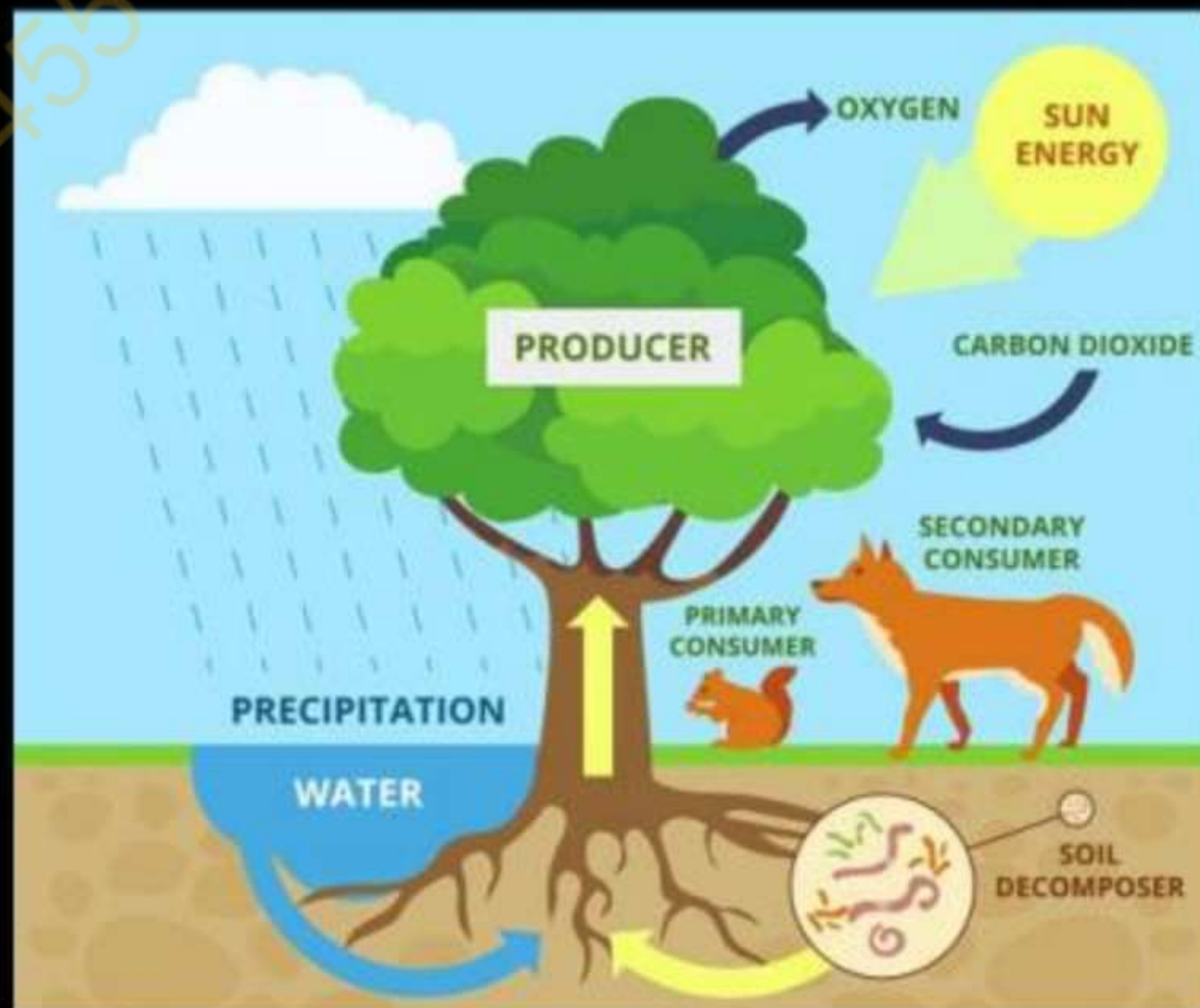
→ They are also called as **tertiary consumers**.

→ **Examples** are human beings and dog.

STRUCTURE OF ECOSYSTEM

(C) Decomposers (अपघटक)

- Those organisms which feeds on dead bodies of consumers and decompose the complex organic compounds into simple organic compounds and make them available for autotrophs(producers).
- They includes mainly bacteria and fungi.



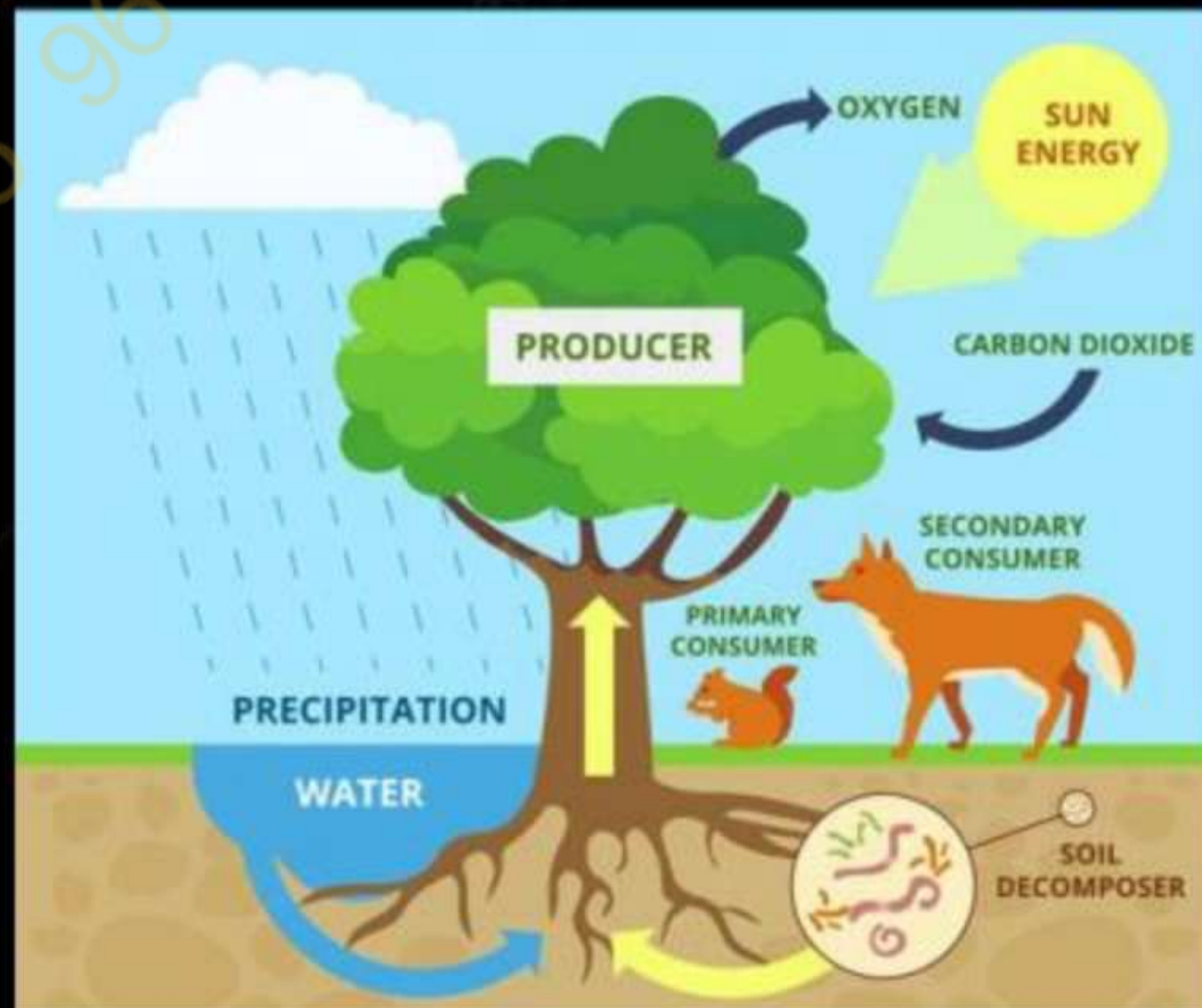
FUNCTIONS OF ECOSYSTEM

(पारिस्थितिकी तंत्र के कार्य)

→ The functions of an ecosystem are essential for maintaining the balance of the environment and supporting life.

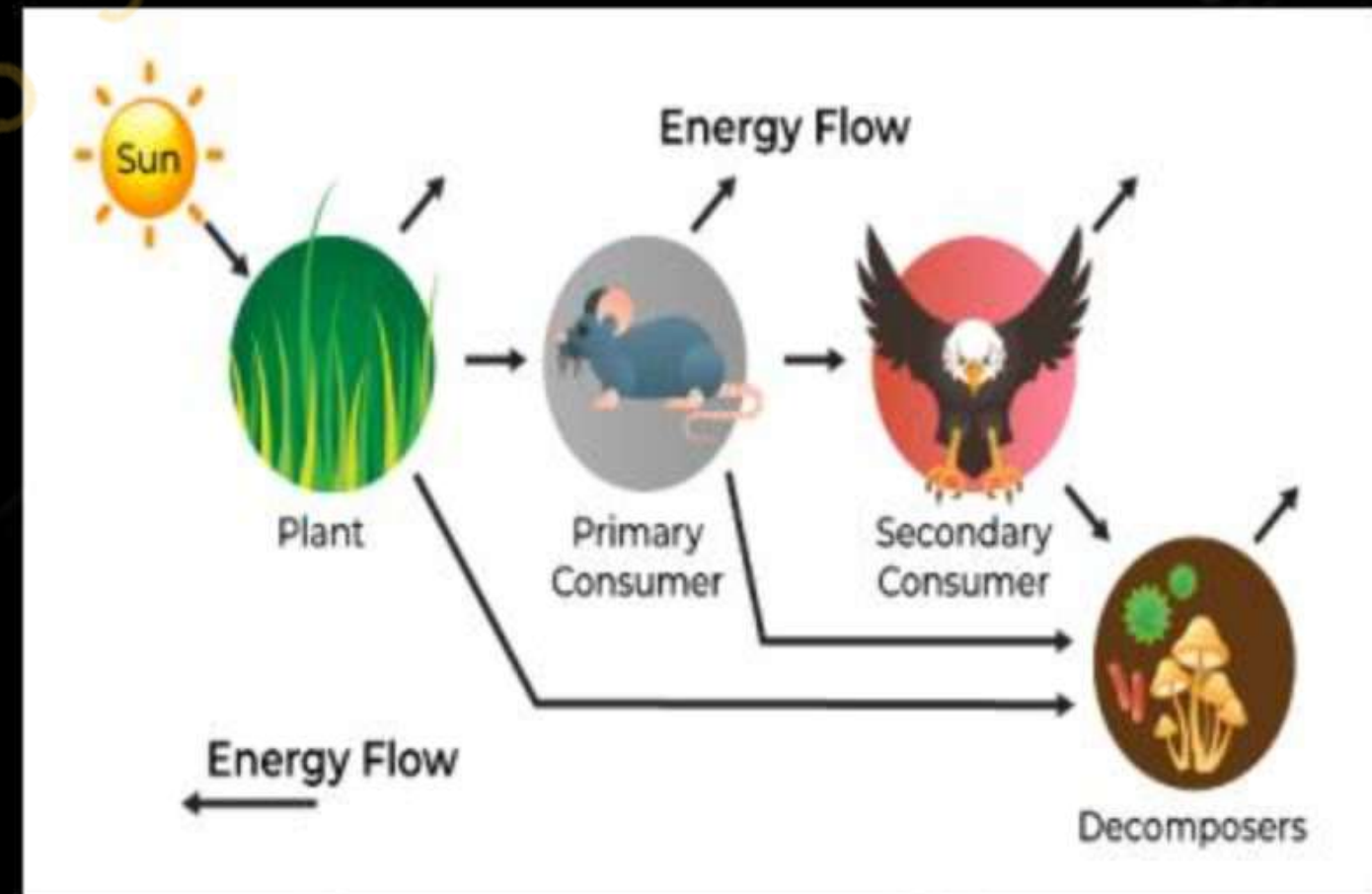
These are the following functions of ecosystem-

- Energy Flow
- Nutrient Cycling
- Regulation of Ecological Processes (climate, water purification, and disease control)
- Habitat Provision
- Soil Formation and Maintenance



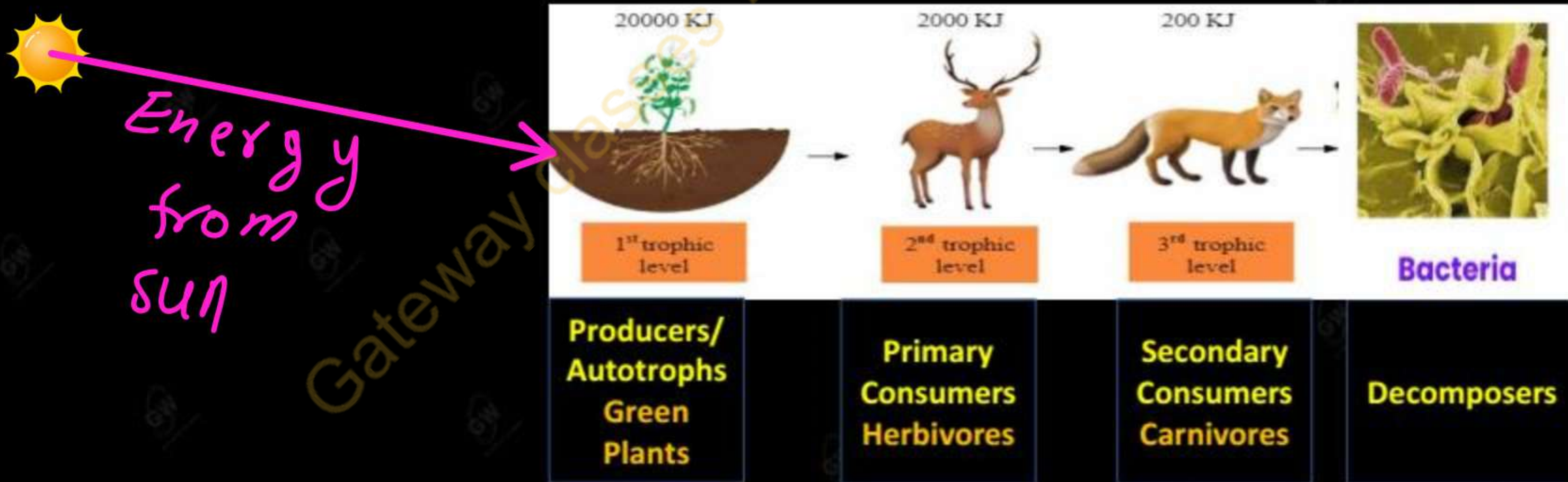
... *Energy Flow in an Ecosystem*

- Energy flow is the movement of energy through an ecosystem through a series of organisms and back to the external environment.
- According to first law of thermodynamics, energy can neither be created nor destroyed; energy can simply change its form.
- In an ecosystem, autotrophs i.e green plants fix the radiant energy of the sun and converted it into chemical energy.
- This chemical energy is transferred to heterotrophs i.e. consumers when they feed upon autotrophs and finally when heterotrophs die this energy is transferred to decomposers.
- Hence in this process of energy flow, we can say that the first law of thermodynamics is applicable in living organisms.



Energy Flow in an Ecosystem

- According to the second law of thermodynamics, during energy transformation, large part of energy is lost in the form of heat into the atmosphere.
- As shown in fig., there is a successive decrease in energy at higher trophic levels.
- The Ten percent law of energy flow states that when the energy is passed on, from one trophic level to another, only 10% of the energy is passed on.



- The efficiency of primary production is very low, less than 1% of radiant energy available to plants (autotrophs) have been captured and transformed to chemical energy.
- It is clear from the fig. that there is unidirectional flow of energy i.e. energy trapped by green plants from sun does not go back to sun and energy which is transferred from green plants to herbivores does not come back to autotrophs.
- Therefore if we remove sun (source of energy), the ecosystem will collapse.
- During process of energy flow there is a loss of considerable amount of energy as heat into atmosphere i.e. there is a successive decrease in energy as we go to higher trophic level.
- So shorter the food chain, greater will be the available food energy.

- The transfer of food energy from the source in plants (producers) through a series of organisms (herbivores → carnivores → decomposers) with repeated stages of eating and being eaten is known as the food chain.
- In any food chain, energy (in the form of food) flow from producers to primary consumers (herbivores), from primary consumers to secondary consumers (carnivores), from secondary consumers to tertiary consumers (carnivores/omnivores) and so on.
- **The transfer of energy from one trophic level to the next trophic level is called food chain.**

There are **three types** of food chain:

1. Grazing Food Chain
2. Detritus Food Chain
3. Parasitic Food Chain

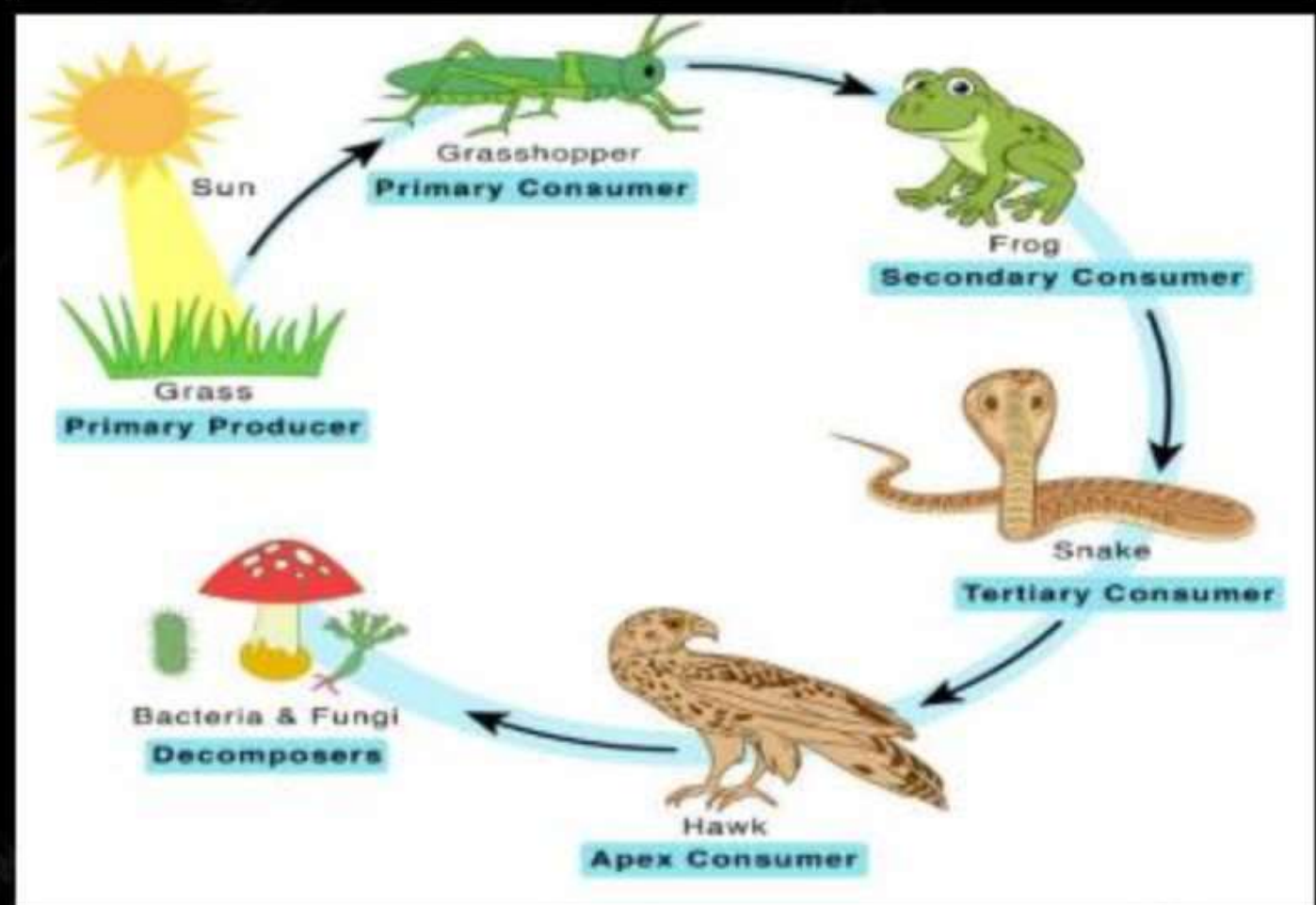
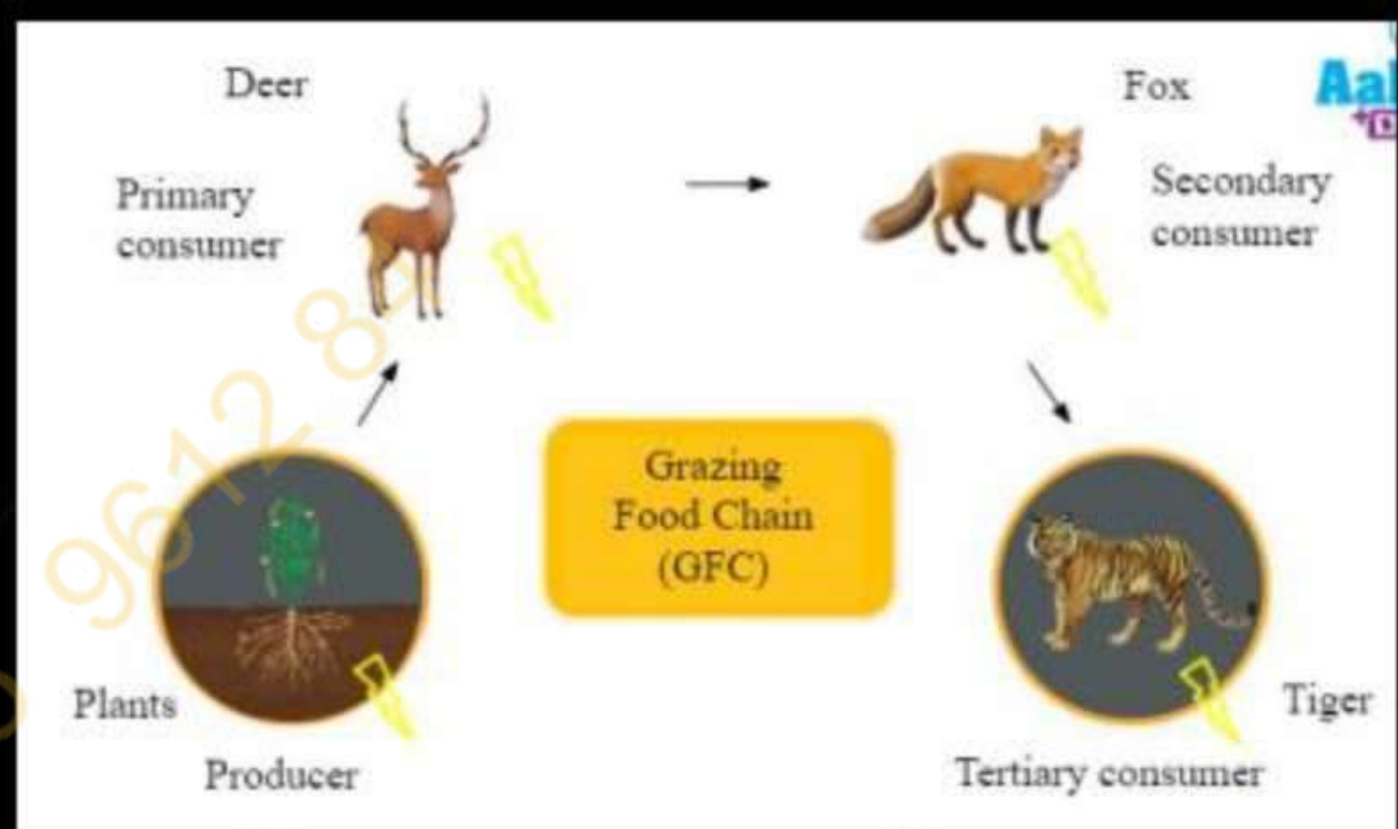


1. Grazing Food Chain

- This type of food chain starts from the living green plants goes to grazing herbivores, and on to carnivores.
- Ecosystems with such type of food chain are directly dependent on an influx of solar radiation.

Examples are

1. Green plants → Deer → Tiger
 2. Grasses → Grasshopper → Frog → Snake → Hawk
- Decomposers ←



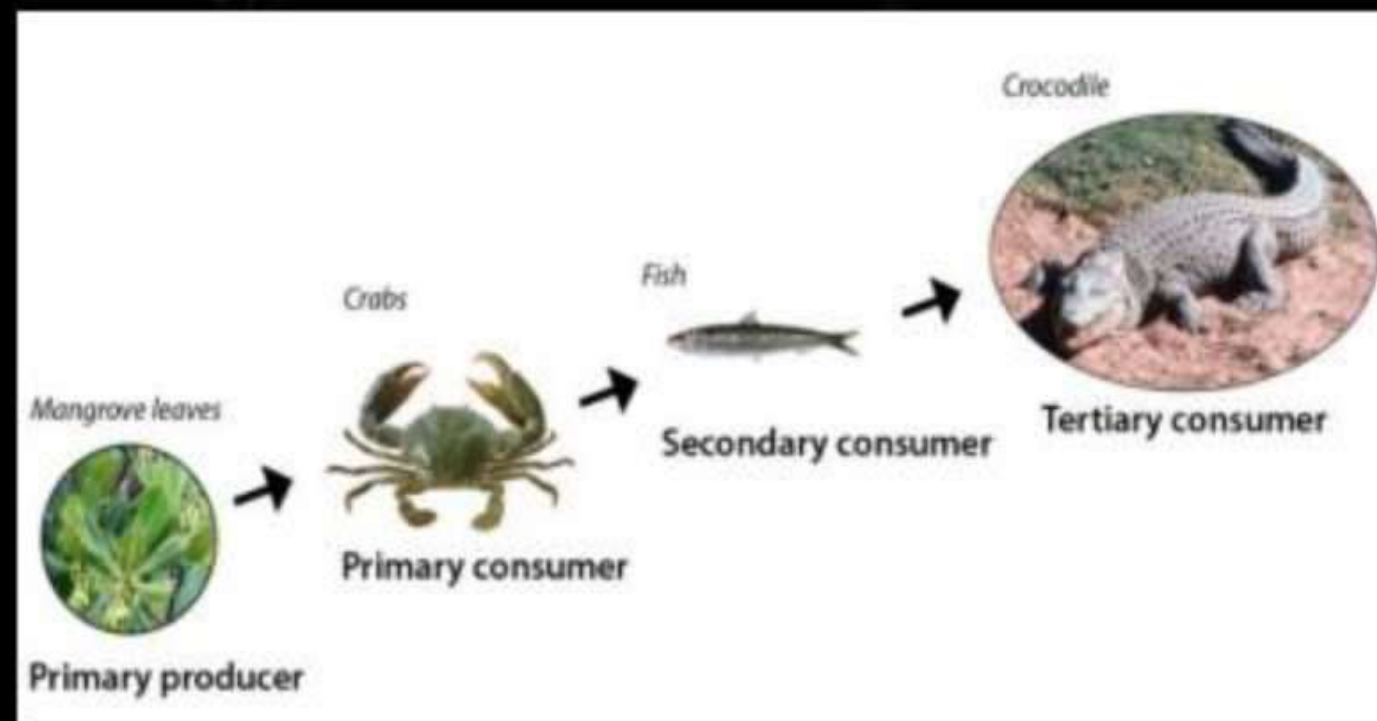
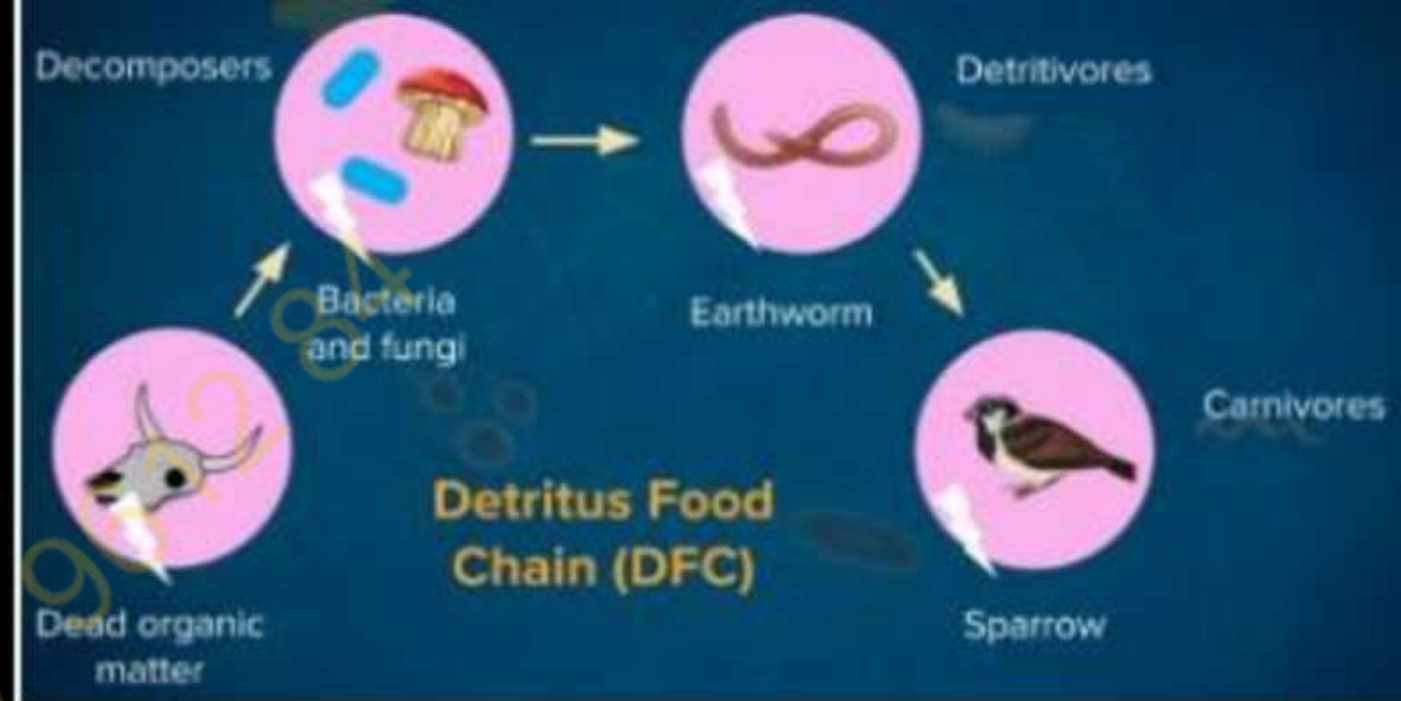
2. Detritus Food Chain

→ It starts from dead organic matter and goes to detritus feeding organisms (detritivores) and on their predators.

→ Detritivores are consumers of dead organic matter.

Examples are

1. Dead organic matter → Detritivores → Predators
2. Mangrove leaves → Crabs → fishes → Crocodile



3. Parasitic Food Chain

- In such type of food chain, parasites may consume a portion of food available to the host.
- In the whole process the parasite gets nutritionally benefited and host is harmed.

Example

Green plants → Sheep → Liver Flukes

Significance of food chain

Energy Transfer: This shows how energy moves through different trophic levels.

Nutrient Cycling: Illustrates the flow of nutrients in ecosystems.

Ecological Balance: Helps maintain stability and balance in ecosystems.

Understanding Ecosystems: Provides insights into species interactions and dependencies.

Conservation Efforts: Aids in identifying critical species and managing ecosystems.

Environmental Health Indicator: Reflects changes and impacts on ecosystems.

FOOD WEB (खाद्य जाल)

AKTU 2022-23

→ "A food web is a tool that illustrates the feeding relationships among species within a specific habitat.

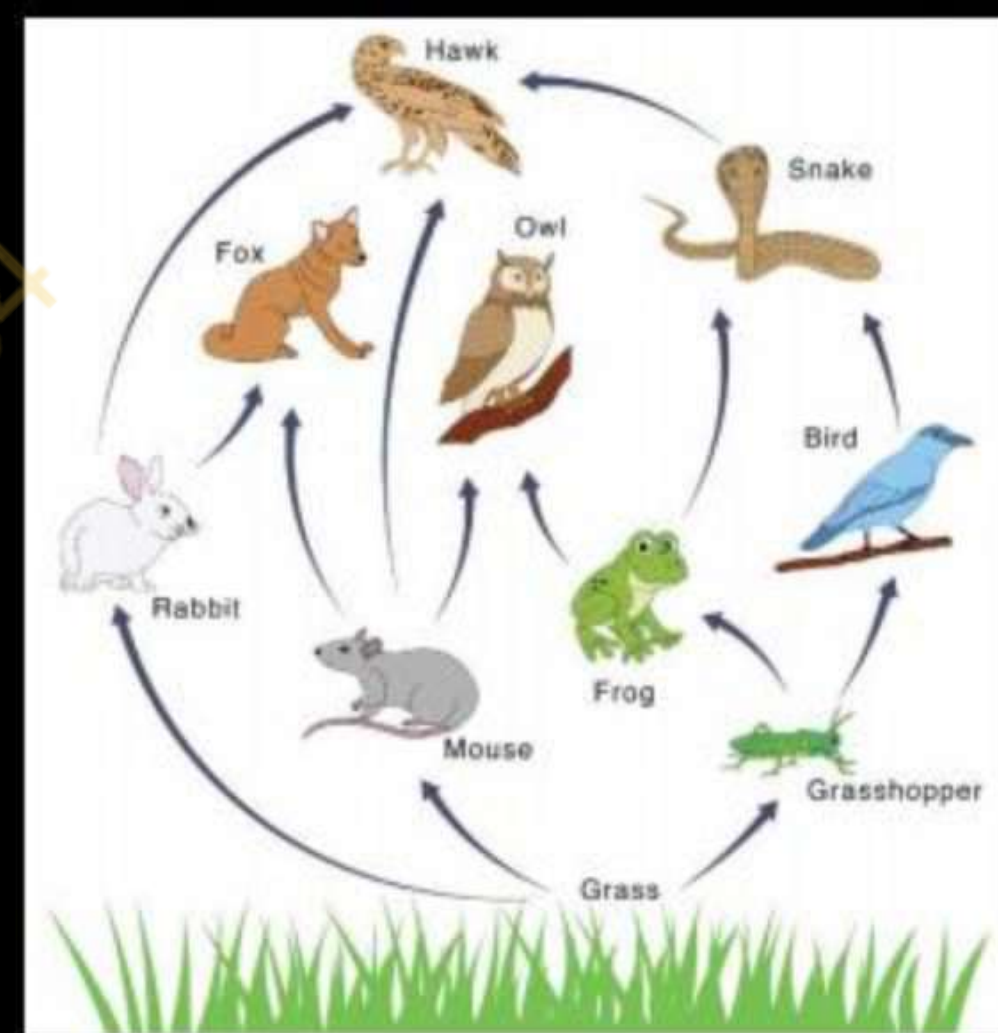
→ Species may occupy different trophic levels in an ecosystem.

→ **For example**, a single plant may be eaten by more than one species for their nourishment,, herbivores feed on different kinds of plants. Similarly, carnivores may feed on different kinds of herbivores, and so on.

→ Therefore, we can say that food chains are interconnected in various ways.

→ **Such types of interconnected food chains form a web called a food web."**

→ "A network of food chains which are interconnected at various trophic levels, so as to form a number of feeding connections amongst different organisms of a biotic community is called food web".



1. Pyramid of Numbers

→ There is a gradual decrease in number of individuals from lower trophic level to higher trophic levels.

→ In a forest or grassland ecosystem autotrophs support a lesser number of herbivores which support fewer carnivores.

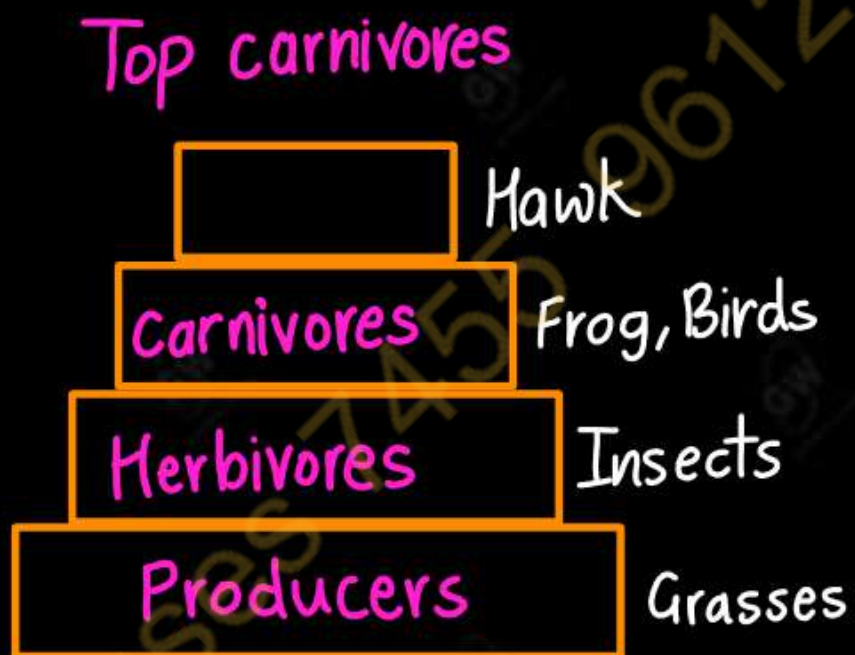


Fig. Grassland

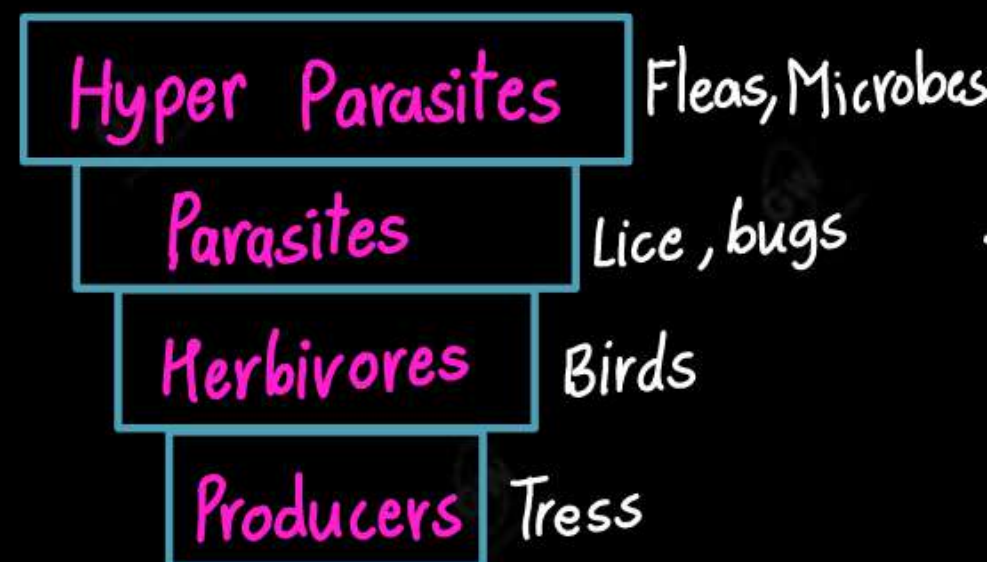


Fig. Parasite food chain

→ In case of parasitic food chain, single tree may support variety of herbivores and these herbivores support various parasites.

→ These parasites may support varieties of hyper parasites like bacteria and fungi.

→ Therefore in **parasitic or inverted pyramid** there is a gradual increase in the number as we go up from lower to higher trophic level.

2. Pyramid of Biomass

- In the pyramid of biomass, there is a gradual **decrease** in biomass from lower to higher trophic level i.e a decrease in biomass from producers to primary consumers and so on.
- In other words the maximum biomass may be formed in producers and the minimum in tertiary as top consumers.

→ In an inverted pyramid of biomass, there is a gradual **increase** in the biomass as we go up from lower to higher trophic levels.

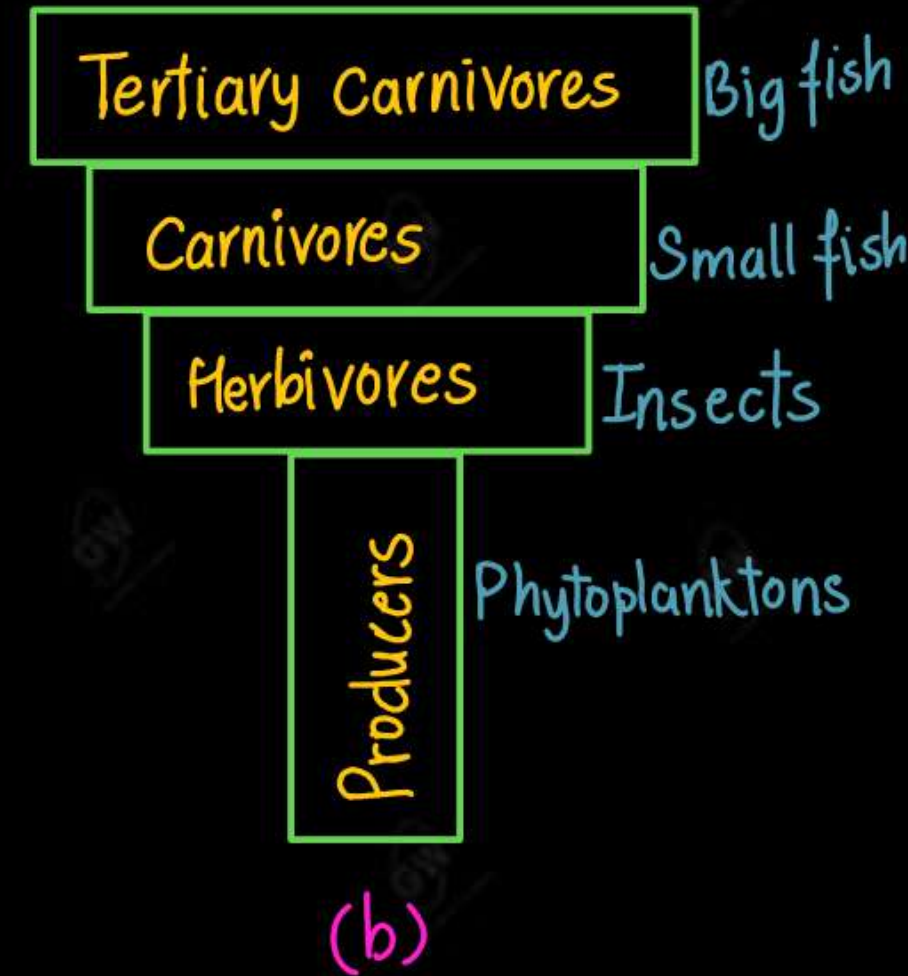
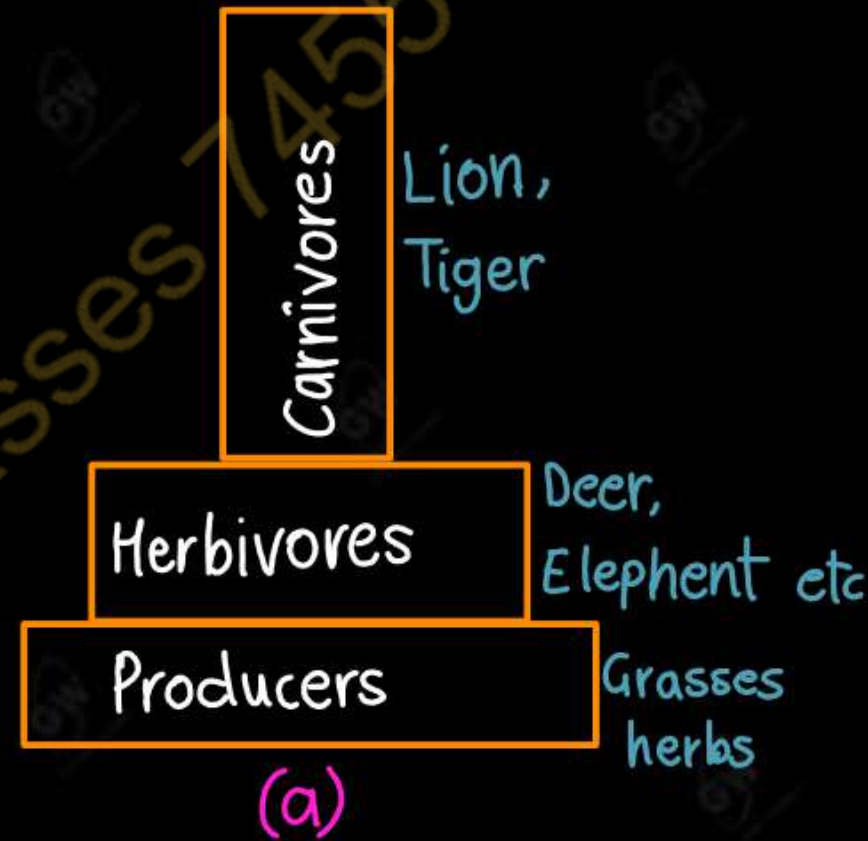


Fig. Pyramid of biomass (a) Forest
(b) Pond

3. Pyramid of Energy

- In pyramid of energy, there is a gradual **decrease** in energy as we go up from lower trophic level to higher trophic level.
- According to second law of thermodynamics there is considerable loss of energy in the form of heat as we go up higher in trophic level.
- More energy is available in autotrophs in comparison to primary consumers.
- Therefore we can easily conclude that shorter the food chain higher will be the amount of energy at top.
- The pyramid of energy is always upright due to unidirectional flow of energy in food chain.



Fig. Pyramid of energy

Why are there fewer Organisms at the Top of the Ecological Pyramid?

1. Energy Loss at Each Trophic Level
2. Smaller Energy Base
3. Greater Energy Needs of Higher Trophic Levels

Importance of Ecological Pyramid

The importance of the ecological pyramid can be explained in the following points:

Energy Flow: Illustrates the decrease in energy as it moves up trophic levels.

Ecosystem Structure: Visual representation of trophic levels and species interactions.

Biomass Distribution: Shows the amount of biomass at each trophic level.

Population Dynamics: Indicates population sizes and trends of different species.

Conservation and Management: Helps identify at-risk species and manage ecosystems sustainably.

Nutrient Cycling: Depicts how nutrients move through an ecosystem.

Ecological Efficiency: Demonstrates the efficiency of energy transfer in ecosystems.

Limitations of ecological pyramid

Ecological pyramids are graphical representations showing the relationship between different trophic levels in an ecosystem. While they are useful for understanding the flow of energy, biomass, and numbers, they have several limitations:

Does Not Consider Decomposers:

→ Ecological pyramids typically do not include decomposers, which play a crucial role in nutrient cycling.

Ignores Species Complexity:

→ They oversimplify the relationships by not accounting for species that belong to multiple trophic levels (omnivores).

No Energy Flow Details:

→ Pyramids do not show the energy transfer efficiency between trophic levels.

Does Not Reflect Nutrient Cycling:

→ They focus on energy and biomass but do not represent the recycling of nutrients within the ecosystem.

- **By 2050**, our planet will need to support some **ten billion** people compared to about **eight billion** today. → This raises huge challenges in **food, clothing, and shelter** for this rapidly growing population.
- Development has to be visualized in a holistic manner, where it brings benefits to all, not only for the present generation but also for future generations.

“The development that meets the needs of the present without compromising the ability of future generations to meet their own needs”.

- Sustainable development emphasizes on a balance between economic growth, environmental protection, and social equity.

Elements of Sustainable Development

1. Economic Sustainability:

- Ensuring economic growth and development that creates wealth and employment, without depleting natural resources.
- It focuses on improving the quality of life and reducing poverty, while maintaining economic activities over time.

2. Environmental Sustainability:

- Protecting and managing natural resources and ecosystems to ensure that they can sustain future generations.
- It involves reducing pollution, conserving biodiversity, and using resources efficiently to prevent degradation of the environment.

3. Social Sustainability:

- Promoting social equity, justice, and access to basic human rights and services.
- It ensures that development benefits all segments of society, including marginalized and disadvantaged groups, fostering inclusive communities

4. Cultural Sustainability:

- Respecting and preserving cultural diversity and heritage.
- This ensures that development respects local customs, traditions, and values, while also promoting cultural exchange and mutual understanding.

The Sustainable Development Goals (SDGs) are a set of 17 global goals established by the United Nations in 2015, designed to be a blueprint for achieving a better and more sustainable future for all.

→ They address global challenges, including those related to poverty, inequality, climate change, environmental degradation, peace, and justice.

These are the following 17 SDGs:

1. No Poverty

2. Zero Hunger

3. Good Health and Well-Being

4. Quality Education

5. Gender Equality

6. Clean Water and Sanitation

7. Affordable and Clean Energy

8. Decent Work and Economic Growth

9. Industry, Innovation, and Infrastructure

10. Reduced Inequality

11. Sustainable Cities and Communities

12. Responsible Consumption and Production

13. Climate Action

14. Life Below Water

15. Life on Land

16. Peace, Justice, and Strong Institutions

17. Partnerships for the Goals

To achieve sustainable development, following measures can be taken:

1.Resource Efficiency:

→ Promote the efficient use of resources, including water, energy, and raw materials, to reduce waste and lower environmental impact.

2.Renewable Energy:

→ Invest in renewable energy sources such as solar, wind, and hydroelectric power to reduce dependence on fossil fuels and lower carbon emissions.

3.Sustainable Agriculture:

→ Adopt farming practices that conserve soil, water, and biodiversity, such as crop rotation, organic farming, and agroforestry.

4.Pollution Control:

→ Implement strict regulations to control pollution, improve waste management, and promote recycling and reuse of materials.

5. Conservation of Biodiversity:

→ Protect natural habitats and endangered species through conservation programs, protected areas, and sustainable land use practices.

6. Public Awareness and Education:

→ Raise awareness about the importance of sustainability and educate the public on environmentally friendly practices and lifestyles.

7. Sustainable Urban Planning:

→ Design cities and communities to be energy-efficient, reduce waste, promote green spaces, and improve public transportation.

8. Global Cooperation:

→ Foster international collaboration to address global environmental challenges such as climate change, biodiversity loss, and pollution.

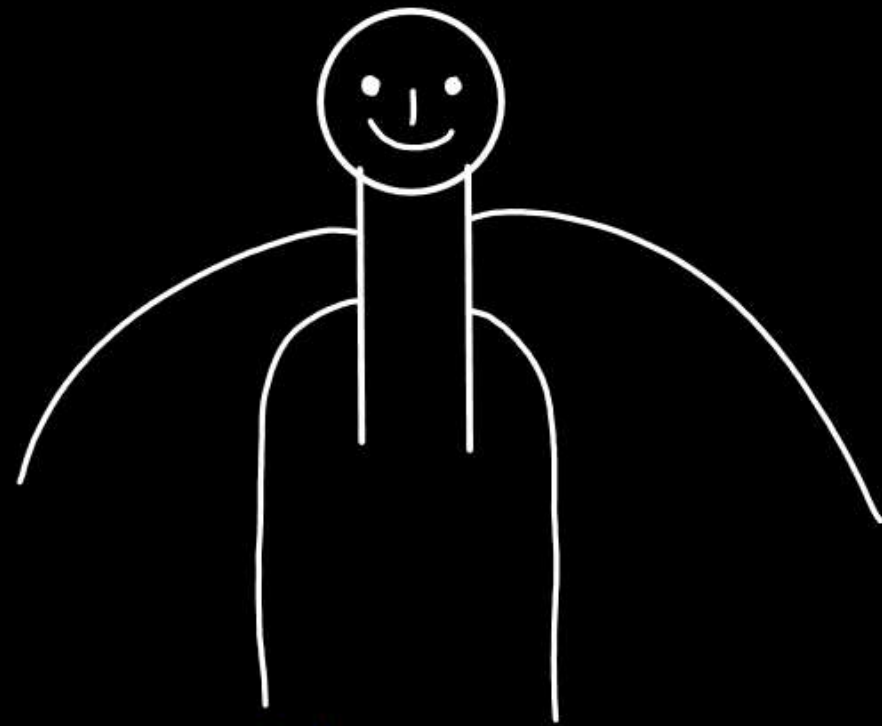
Reasons for Environmental Crisis

- Overpopulation
- Industrialization
- Urbanization
- Deforestation
- Pollution
- Climate Change
- Overconsumption

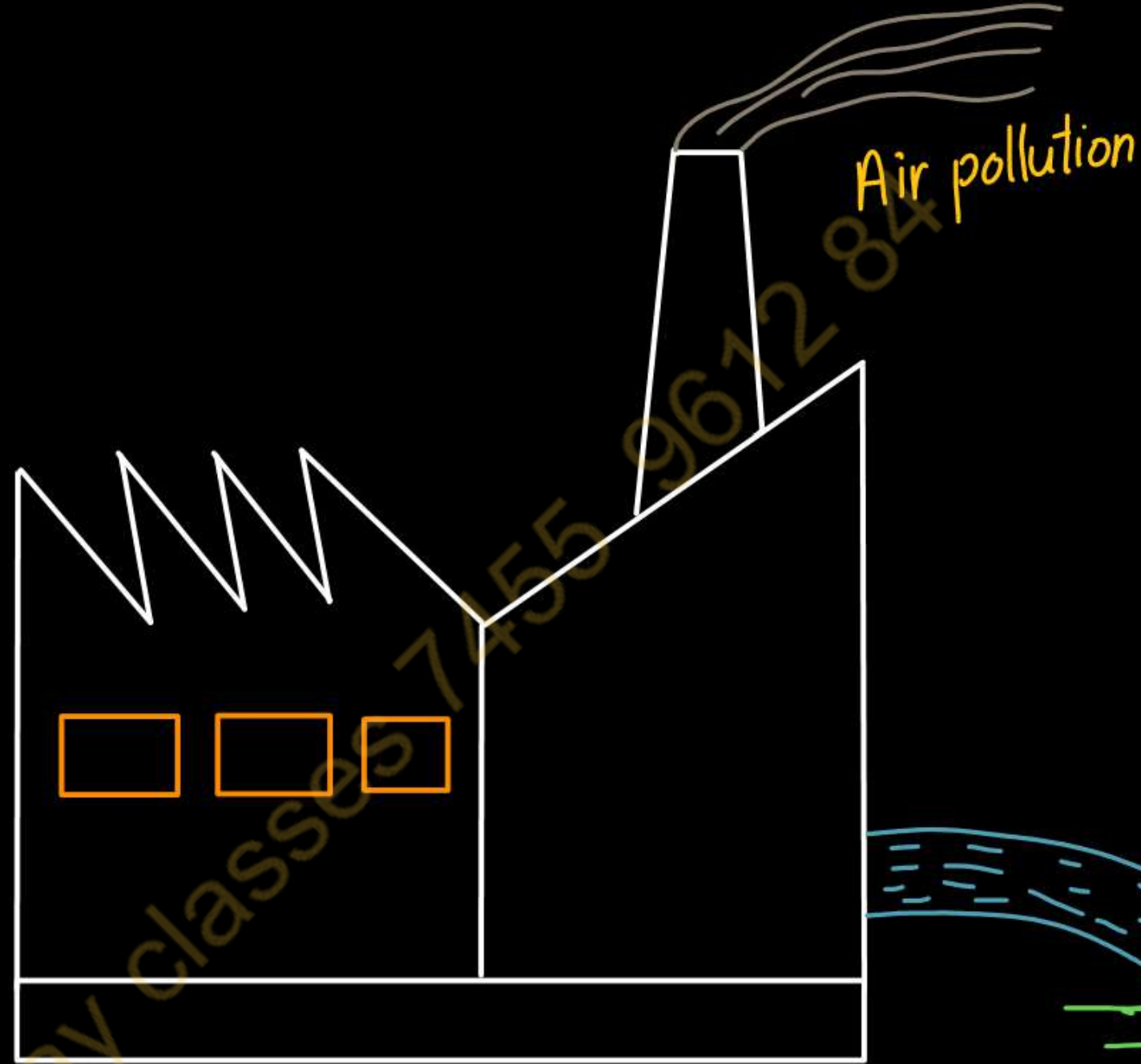
- It is an important management tool for ensuring optimal use of natural resources for sustainable development.
- It is a process of evaluating the environmental impacts of a proposed project or development taking inter-related socio-economic, cultural, and human-health impacts, **both beneficial and adverse.**

United Nations Environment Program (UNEP) defines Environmental Impact Assessment (EIA) as a tool used to identify the environmental, social and economic impacts of a project prior to decision-making.

- It aims to predict environmental impacts at an early stage in project planning and design, find ways and means to reduce adverse impacts, shape projects to suit the local environment and present the predictions and options to decision-makers



Voilation
Env. Norms or not



Air pollution

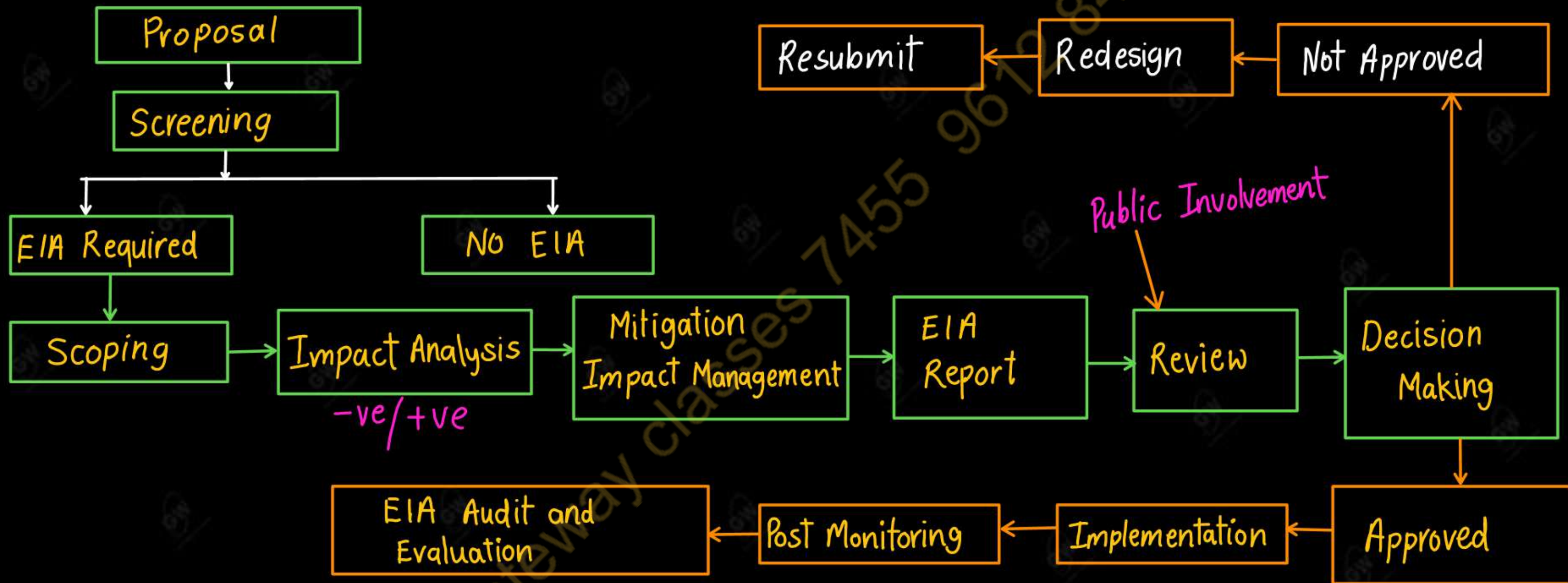
Chemical Industry

water
body

NOC

ENVIRONMENTAL IMPACT ASSESSMENT-Flow Chart

(पर्यावरणीय प्रभाव आकलन)



Steps in EIA process –

1. Project Screening - This entails the application of EIA to those projects that may have significant environmental impacts. It is quite likely, however, that screening is done partly by the EIA regulations, operating in a country at the time of assessment. Screening Criteria are based upon:

- Scales of investment
- Type of development
- Location of development

2. Scoping - This step seeks to identify, at an early stage, the key significant environmental issues from among a host of possible impacts of a project and all the available alternatives.

- Baseline data describes the existing environmental status of the identified study area.
- The site-specific primary data should be monitored for the identified parameters and supplemented by secondary data if available.
- The primary data are those, which need to be collected in the field to define the status of the environment (like air quality data, water quality data etc).
- The secondary data are those data, which have been collected over the years and can be used to understand the existing environmental scenario of the study area.

4. Impact Prediction –

- Identify the project activities, which may have an impact on the environment during the planning, construction, and operational phases of a new activity.
- Positive and negative, reversible and irreversible, and temporary and permanent impacts need to be predicted which presupposes a good understanding of the project by the assessment agency.

The following impacts of the project should be assessed: Air, Noise, Water, Land, Biological, Socio-Economic

5. Assessment of Alternatives, Delineation of Mitigation Measures and Environmental Impact Assessment Report

- For every project, possible alternatives should be identified and environmental attributes compared.
- Alternatives should cover both project location and process technologies.
- Alternatives should then be ranked for selection of the best environmental option for optimum economic benefits to the community at large.
- Once alternatives have been reviewed, a mitigation plan should be drawn up for the selected option and is supplemented with an Environmental Management Plan (EMP) to guide the proponent towards environmental improvements.
- An EIA report should provide clear information to the decision-maker on the different environmental scenarios without the project, with the project and with project alternatives.

6. Public Hearing

- Law requires that the public must be informed and consulted on a proposed development after the completion of EIA report.
- Any one likely to be affected by the proposed project is entitled to have access to the Executive Summary of the EIA.
- The affected persons may include: bonafide local residents; local associations; environmental groups active in the area; any other person located at the project site/ sites of displacement.
- They are to be given an opportunity to make oral/ written suggestions to the State Pollution Control Board.

7. Decision Making - Decision making process involve consultation between the project proponent (assisted by a consultant) and the impact assessment authority (assisted by an expert group if necessary). The decision on environmental clearance is arrived through a number of steps including evaluation of EIA and EMP.

8. Monitoring the Clearance Conditions –

- Monitoring should be done during both the construction and operation phases is not only to ensure that the commitments made are completed with but also to observe whether the predictions made in the EIA reports were correct or not.
- Where the impacts exceed the predicted levels, corrective action should be taken.
- Monitoring will enable the regulatory agency to review the validity of predictions and the conditions of implementation of the Environmental Management Plan (EMP).

Importance of EIA

- EIA links environment with development for environmentally safe and sustainable development.
- EIA provides a cost effective method to eliminate or minimize the adverse impact of developmental projects.
- EIA enables the decision makers to analyze the effect of developmental activities on the environment well before the developmental project is implemented.
- EIA encourages the adaptation of mitigation strategies in the developmental plan.
- EIA makes sure that the developmental plan is environmentally sound and within the limits of the capacity of assimilation and regeneration of the ecosystem.



Thank You

THE EFFECTS OF HUMAN ACTIVITIES ON ENVIRONMENT

- The population and activities of every species depend on the resources available to them and interaction among species.
- The waste product of one species may become the food for another species.
- Humans alone have the potential to gather resources from their surroundings and process them into different forms.
- As a result of which the anthropogenic (man-induced) pollutants have overloaded the system, and the natural equilibrium is disturbed

- Primitive humans used natural resources to satisfy their basic needs of air, water, food and shelter.
- These natural and unprocessed resources were readily available in the biosphere, and the residues produced by the use of these resources were generally compatible with or easily assimilated by the environment.
- Technological development has resulted in fast depletion of natural resources.
- Mining activities, urbanization and industrialization have all interfered with the ecological balance of nature due to large scale impact.
- Humans turned their attention to other needs beyond those associated with survival.

- Automobiles, household appliances, processed foods and beverages, etc. have now become as popular as to seem necessities, and meeting these acquired needs had become a major thrust of modern industrial society.
- These acquired needs are usually met by items that must be processed or manufactured or refined.
- The production, distribution and use of such items usually results in more complex residuals and/or wastes, many of which are not compatible with the environment.

Food (Food Security)

Environmental Impacts of Food Production

- By the time the food you eat, much of the environmental impact has already occurred.
- Before food production even starts, natural habitats and ecosystems are destroyed to clear land that will be used for agriculture.

1. Water Use & Water Pollution

- Growing food takes a lot of water. About 70% of all water use goes towards agricultural efforts. When runoff of agricultural pollutants occurs, groundwater get contaminated with things like nitrogen and phosphorus- commonly used in modern farming practices.

2. Greenhouse Gas Emissions

- Greenhouse gas emissions, such as Carbon dioxide, are created when fossil fuels are used during several aspects of the food cycle, including food production and distribution.

3. Environmental Contaminants & Pollutants

- The growing, producing, and transporting of food can generate various contaminants that can have adverse effects on the health of humans and the ecosystem.
- These contaminants include ammonia and the emission of different nitrogen compounds that disrupt the soil, animal and plant life.

4. Depletion of Natural Resources

- Food production takes up a significant portion of the world's natural resources.

5. Degradation of Land

- Harvesting the crop represents a significant amount of nutrients, water, and energy being taken from the land.
- This leaves the land barren, and unfriendly for the growth and development of new organisms and ecosystems.

Threats to food security -

- Declining productivity.
- High dependence on imported food.
- Increasing poverty.
- Declining income from traditional crops.
- Growing food-related diseases.

Gateway classes 7455 961284

Major world food problems - There are two kinds of food insufficiency undernourishment and malnourishment.

1. Undernourishment (कुपोषण)–

- Undernourishment is the lack of sufficient calories in available food, so that one has little or no ability to move or work.
- The Food and Agricultural Organization (FAO) estimates that the average minimum daily calorie intake is about 2500 calories per day.
- People who receive less than 90% of the minimum dietary intake on a long-term basis are considered undernourished.
- Undernourished persons have less energy for doing any kind of work.
- They are susceptible to diseases, their body becomes weak and they frequently fall sick.

2. Malnourishment (कुपोषण)–

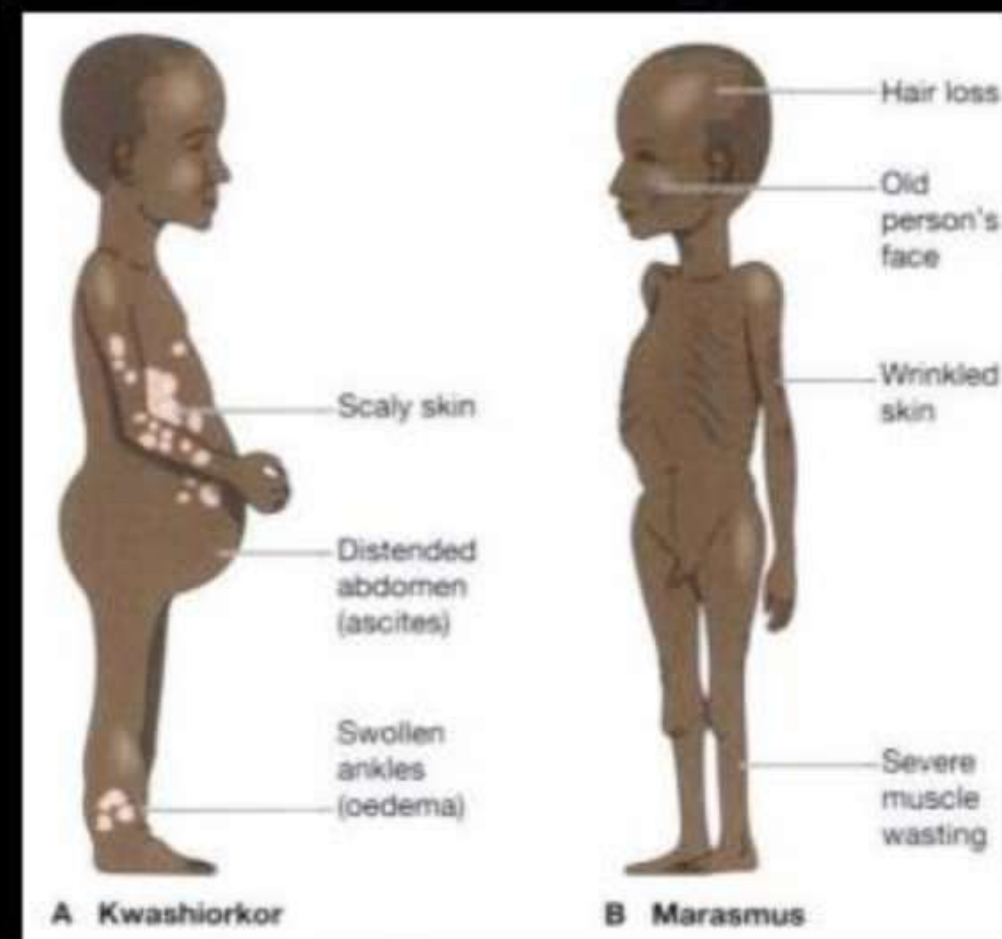
- It means lack of essential nutrients (like proteins, vitamins, fats, minerals etc) in the diet.
- It is possible to have excess food and still suffer from malnourishment due to nutritional imbalance caused by a lack of specific dietary components or an inability to absorb or utilize essential nutrients. Some of the major problems of malnourishment are -

(i) Marasmus –

→ Marasmus is a kind of malnutrition that occurs **due to deficiency of proteins, carbohydrates and fats in the diet.**

→ In marasmus the child consumes little or no food leading to minimal calorie intake and also a total lack of nutrients.

Symptoms are Weight loss, dehydration, slowing of growth, shrinkage of stomach.



→ It is caused by protein deficiency. Symptoms are bulging of abdomen, mental retardation and stunted growth.

(iii) Goitre - An Iodine deficiency in the diet.

3.Over Nourishment

→ It is due to over consumption of fats, sugar and salt in diet. Symptoms are Obesity, High blood pressure, Diabetes, Heart diseases etc.

Balanced diet

→ A balanced diet is one that contains all of the essential elements that the human body needs.

→ Carbohydrates, lipids, vitamins, minerals, proteins, fiber and water are all essential components in a well-balanced diet.

→ A nutritious, well- balanced diet lowers the risk of disease and enhances general health.

Food security can be achieved by -

- Implementing appropriate food, agriculture and rural development policies with emergency and disaster planning.
- Permitting appropriate International Agricultural agreements.
- Minimizing over-exploitation of natural resources that may jeopardize food security in the long run.
- Ensuring that the imported food products are –
 - (i) of acceptable quality and safe to eat.
 - (ii) does not reduce agricultural employment levels.
 - (iii) permit protection of national food security.
- Facilitating proper distribution of sufficient, safe and nutritious food.

Effects of Human Activities on Environment

Agriculture

Effects of Agriculture Activities on Environment

→ Agriculture is the world's oldest and largest industry and more than half of all the people in the world still live on farms.

→ Agriculture has both primary and secondary environmental effects.

A **primary effect** is an effect on the area where the agriculture takes place, i.e. on-site effect.

A **secondary effect**, also called an off-site effect, is an effect on an environment away from the agricultural site.

The effects of agriculture on the environment can be broadly classified into three groups - **Local, Regional and Global.**

i. Local Changes –

- These occur at or near the site of farming.
- These changes/effects include soil erosion and increase in sedimentation downstream in local rivers.
- Fertilizers carried by sediments can cause eutrophication of local water bodies.
- Polluted sediments can also transport toxins and destroy local fisheries.

ii. Regional Changes

- They generally result from the combined effects of farming practices in the same large region.
- Regional effects include deforestation, desertification, large scale pollution, increases in sedimentation in major rivers.

iii. Global Changes - These include climatic changes as well as extensive changes in chemical cycles.

The main impacts of traditional type of agriculture are as follows:

- (i) Deforestation** - The slash and burn of trees in forests to clear the land for cultivation and frequent shifting result in loss of forest cover.
- (ii) Soil Erosion** - Clearing of forest cover exposes the soil to wind, rain and storms, thereby resulting in loss of top fertile layer of soil.
- (iii) Depletion of Nutrients** - During slash and burn the organic matter in the soil gets destroyed and most of the nutrients are taken up by the crops within a short period, thus making the soil nutrient poor which forces the cultivators shift to another area.

Effect of Modern Agriculture Practices on the Environment

→ The major problems that have arisen due to modern agricultural practices are

i. Fertilizer Related Problems:

- (a) Micronutrient Imbalance
- (b) Nitrate Pollution
- (c) Eutrophication

ii. Pesticides-Related Problems

iii. Water Logging

iv. Salinity

Gateway classes 7455 9612 84

Pesticides

→ A pesticide is a chemical substance used to kill or control pests, such as insects, weeds, fungi, and rodents. Pesticides help protect crops from damage and increase agricultural productivity.

Pesticides-Related Problems

1. Harm to Other Animals

- Pesticides intended to kill pests can also harm other animals like birds, and fish.
- This reduces the population of beneficial insects (like pollinators) and other wildlife.

2. Health Risks for People

- People can get exposed to pesticides by breathing them in, touching them, or eating food with pesticide residues.
- This exposure can cause immediate health issues like poisoning or long-term problems like cancer and neurological disorders.

3. Environmental Pollution:

- Pesticides can seep into the soil and water bodies.
- This pollution can harm plants and animals living in these environments and disrupt ecosystems.

4. Build-Up of Toxins:

- Pesticides can accumulate in the bodies of animals and become more concentrated as they move up the food chain.
- This can lead to high levels of toxins in larger animals and humans who consume them, affecting health and biodiversity.

Effects of Human Activities on the Environment: Transportation

→ Transportation is a major human activity that significantly impacts the environment.

Air Pollution:

Cause: Emissions from cars, trucks, airplanes, and ships.

Effect: Release of harmful pollutants like carbon monoxide, nitrogen oxides, and particulate matter, which contribute to air quality degradation and health problems like asthma and lung disease.

Greenhouse Gas Emissions:

Cause: Burning of fossil fuels for transportation.

Effect: Increased levels of carbon dioxide (CO₂) and other greenhouse gases in the atmosphere, contributing to global warming and climate change.

Noise Pollution:

Cause: Traffic, aircraft, and other transportation-related activities.

Effect: Disturbance to wildlife and humans, leading to stress, hearing loss, and reduced quality of life.

Habitat Destruction:

Cause: Construction of roads, highways, airports, and railways.

Effect: Loss of natural habitats for animals and plants, leading to reduced biodiversity.

Water Pollution:

Cause: Runoff from roads and highways, and oil spills from ships.

Effect: Contamination of water bodies, harming aquatic life and affecting water quality.

Land Use:

Cause: Expansion of infrastructure for transportation.

Effect: Reduction of green spaces and agricultural land, impacting ecosystems and local communities.

Effects of Human Activities on the Environment: Housing

→ Housing development has a significant impact on the environment.

Deforestation and Habitat Destruction:

Cause: Clearing land for building homes and communities.

Effect: Loss of forests and natural habitats, reducing biodiversity and disrupting ecosystems.

Resource Consumption:

Cause: Use of materials like wood, concrete, and metals in construction.

Effect: Depletion of natural resources and increased environmental degradation from mining and logging.

Energy Use:

Cause: Heating, cooling, lighting, and powering homes.

Effect: Increased demand for energy, often generated from fossil fuels, leading to higher greenhouse gas emissions.

Water Consumption:

Cause: Daily household use, landscaping, and construction processes.

Effect: Increased pressure on local water supplies and potential for water shortages.

Waste Generation:

Cause: Construction, household waste, and disposal of old materials.

Effect: Increased landfill use, pollution from improper disposal, and greater need for waste management systems.

Pollution:

Cause: Use of chemicals in construction and household products.

Effect: Release of pollutants into air, water, and soil, affecting health and ecosystems.

SHELTER SECURITY

- Shelter refers to a basic structure or building that covers or provides protection.
- It also includes protection from the weather and places of refuge such as emergency shelter or victims of natural disaster or wars.
- Homelessness is the condition and social category of people who lack housing because they cannot afford or otherwise unable to maintain regular, safe and adequate shelter.

Causes of homelessness - People become homeless for following reasons

- Unemployment.
- Lack of money to buy a property or to pay rent.
- Natural disaster or war
- Family breakdown due to domestic violence.
- Increased wealth disparity and income inequality leading to distortions in the prices of houses.
- Mortgage disclosure of homes by banks.

ECONOMICS SECURITY

- Economic Security refers to the condition of **having stable income** to support a standard of living now and in the future.
- When Children, Adults and Families are safe, healthy, educated and have sufficient money for comfortable living, it means they are economically secure.
- For a country to be developed, the basic requirement is economic security of the citizens.
- Unemployment is the major hindrance in the development of a country.
- It is the state in which a person is without work, unable to work and is currently seeking work.

SOCIAL SECURITY

→ Social Security refers to social welfare services concerned with social protection or protection against socially recognized conditions including poverty, old age, disability, unemployment and other.

Social security may refer to -

Social insurance - In social insurance people receive benefits or services in recognition of contribution to an insurance scheme. These services include provision for retirement pensions, disability insurance, survivor benefits, and unemployment insurance.

Income maintenance - It includes the distribution of cash in the event of interruption of employment including retirement, disability and unemployment.

Services - It refers to services provided by administrations responsible for social security. In different countries this may include medical care, aspects of social work and even industrial relations.

Basic security - Social security is also used to refer to basic security, a term equivalent to access to basic necessities such as food, clothing, shelter, education and medical care.



Thank You